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NOVEMBER 2026

# The 2026 Pennsylvania Talent Profile

Emerson Johnston
Amy Zegart

The bright caps, **54 in every 100**, mark the international share of Pennsylvania's STEM master's degrees.



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By Emerson Johnston and Amy Zegart

**NOVEMBER 2026**

A publication jointly produced by the Hoover Institution and the Institute for Human-Centered Artificial Intelligence at Stanford University.





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Data come from the Integrated Postsecondary Education Data System (IPEDS), the Department of Homeland Security's SEVIS by the Numbers, the National Assessment of Educational Progress, NSF's Survey of Graduate Students and Postdoctorates and Survey of Earned Doctorates, the Department of Labor's Office of Foreign Labor Certification, the USCIS H-1B Employer Data Hub, the Census Bureau's American Community Survey and USPTO patent data via PatentsView, processed in Python; every figure and table is generated from those sources, and each one names its source.

# About This Profile

This profile is a companion to *The 2026 US Talent Report: Benchmarking America's STEM Pipeline Amid the Great Talent Competition*. It applies the report's measures to Pennsylvania: the students the state develops at home, those it draws from abroad, and how both compare with the nation and with the states closest to it in size.

Degrees, enrollment, prices, outcomes and faculty come from IPEDS, the federal census of U.S. colleges and universities, harmonized across 1980–2025. Countries of citizenship come from SEVIS by the Numbers, the Department of Homeland Security's monthly count of active F-1 and M-1 students. Each figure names its source. Peer states are the four closest in total fall enrollment.



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# How to Cite this Profile

Emerson V. Johnston and Amy Zegart, “The 2026 Pennsylvania Talent Profile,” Technology Policy Accelerator, Hoover Institution, and the Institute for Human-Centered Artificial Intelligence, Stanford University, Stanford, CA, November 2026.

# Public Data and Tools

The data behind every chart come from the Integrated Postsecondary Education Data System (IPEDS), published by the National Center for Education Statistics, and from SEVIS by the Numbers, published by the Department of Homeland Security's Student and Exchange Visitor Program. An interactive version of this profile accompanies the report.

To share any charts and figures, proper attribution is required. This profile is licensed under Attribution-NoDerivatives 4.0 International (CC BY-ND 4.0). Please cite as: The 2026 Pennsylvania Talent Profile.

AT A GLANCE

# What Pennsylvania's numbers say

1. **International students earn 54% of Pennsylvania's STEM master's degrees,** 1.4 points above the U.S. (52%) in 2025, ranking 14th of 51 on dependence. The share is 45% for STEM research doctorates (U.S. 43%) and 8.3% for STEM bachelor's (U.S. 6.6%).
2. **Pennsylvania awarded 5.51K STEM master's degrees to international students in 2025,** 4.7% of the U.S. total and 7th of 51 among states. Since 2015 the count changed +78%, against 2.2× nationally.
3. **The heaviest dependence is in engineering technology,** where international students earned 68% of master's degrees (336 degrees; U.S. 54%). 5 STEM fields with at least 50 master's graduates are majority-international.
4. **Among states of similar size,** Pennsylvania ranks 2nd of 5 on the international share of STEM master's: Pennsylvania 54%, Ohio 58%, Georgia 44%, Arizona 41%, North Carolina 49%.
5. **68% of Pennsylvania's first-time students in fall 2024 were state residents** and 3.4% came from abroad. More students arrived from other states (29.9K) than residents left (22.3K), a net gain of 7.58K; the top destination for those leaving was New York.

**54%**

of the STEM master's degrees Pennsylvania awards go to international students (U.S. 52%).





SCORECARD

## Nine numbers against the nation

**637K** #7
**Students enrolled, 2024**
U.S. **19.7M** · −16% since 2015

**5.51K** #7
**International STEM master's, 2025**
U.S. **117K** · +78% since 2015

**54%** #14
**International share of STEM master's, 2025**
U.S. **52%** · +6.9 pp since 2015

**18%** #17
**International share of grad students, 2024**
U.S. **17%** · +2.1 pp since 2015

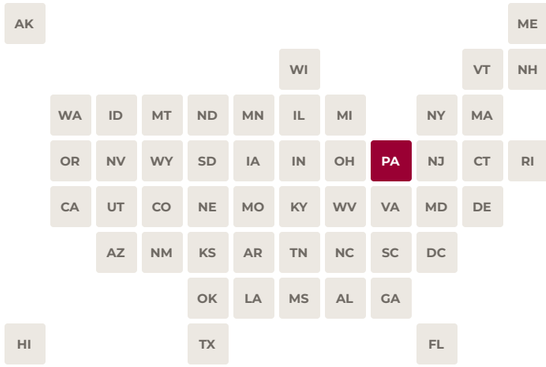
**27%** #10
**STEM share of bachelor's, 2025**
U.S. **24%** · +6.2 pp since 2015

**68%** #15
**Six-year graduation rate, 2024**
U.S. **65%** · +0.9 pp since 2015

**$7.14K** #46
**Appropriations per student, 2024**
U.S. **$11.4K** · +20% since 2015

**68%** #31
**First-time students from in state, 2024**
U.S. **–** · −2.4 pp since 2016

**$109K** #14
**Average faculty salary, 2024**
U.S. **$106K** · −8.2% since 2016

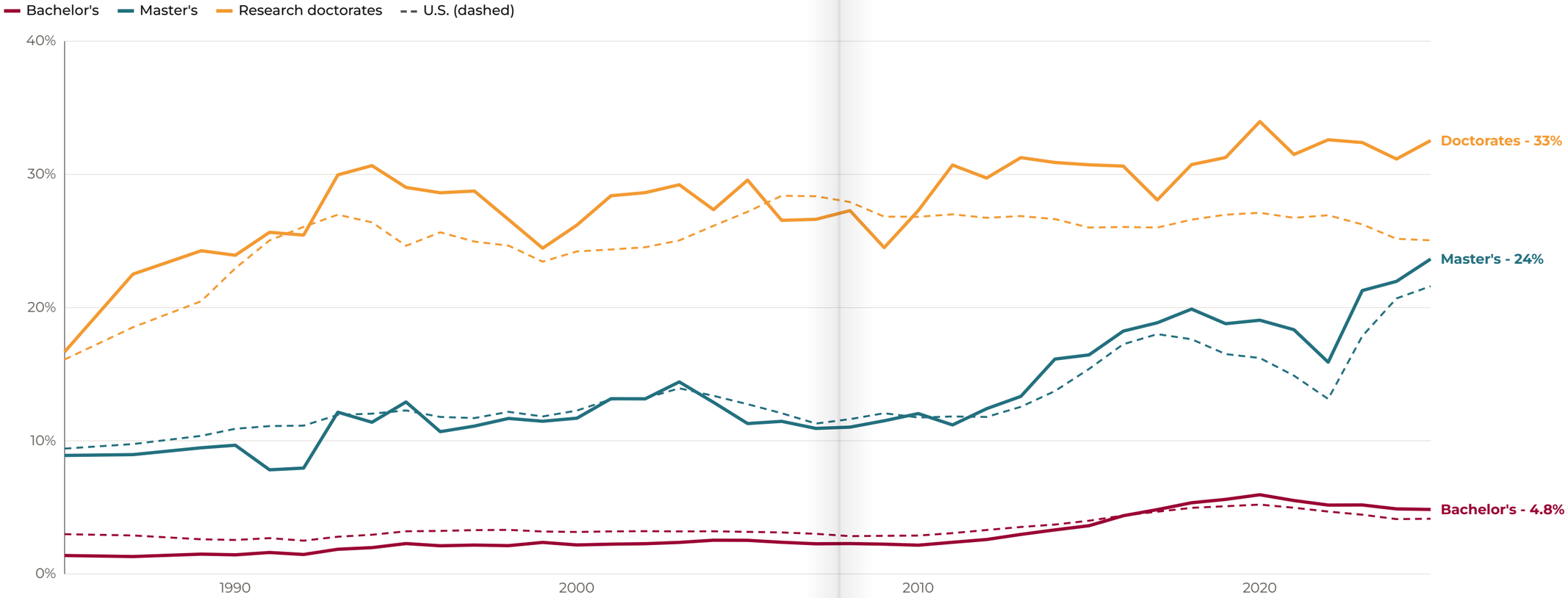



Badges give Pennsylvania's rank among the 50 states and DC (1 = highest). Changes are since 2015.

SECTION 1

# International Dependence

## Figure 1 What Share of Pennsylvania's Degrees Go to International Students?




International share of degrees awarded, all fields; state solid, U.S. dashed. Data from NCES, IPEDS Completions, 1985–2025.

International students make up 18% of Pennsylvania's graduate enrollment and 3.9% of its undergraduates. Dependence climbs with each degree: 4.8% of bachelor's, 24% of master's and 33% of research doctorates go to international students.

In STEM the shares are higher still: 8.3% of STEM bachelor's, 54% of STEM master's and 45% of STEM doctorates. The state awards 4.7% of all U.S. international STEM master's, 7th of 51.

**18%**

**Int. share of grad students, 2024**

U.S. **17%** · **17th** of 51 · +2.1 pp since 2015

**24%**

**Int. share of all master's, 2025**

U.S. **22%** · **12th** of 51 · +7.2 pp since 2015

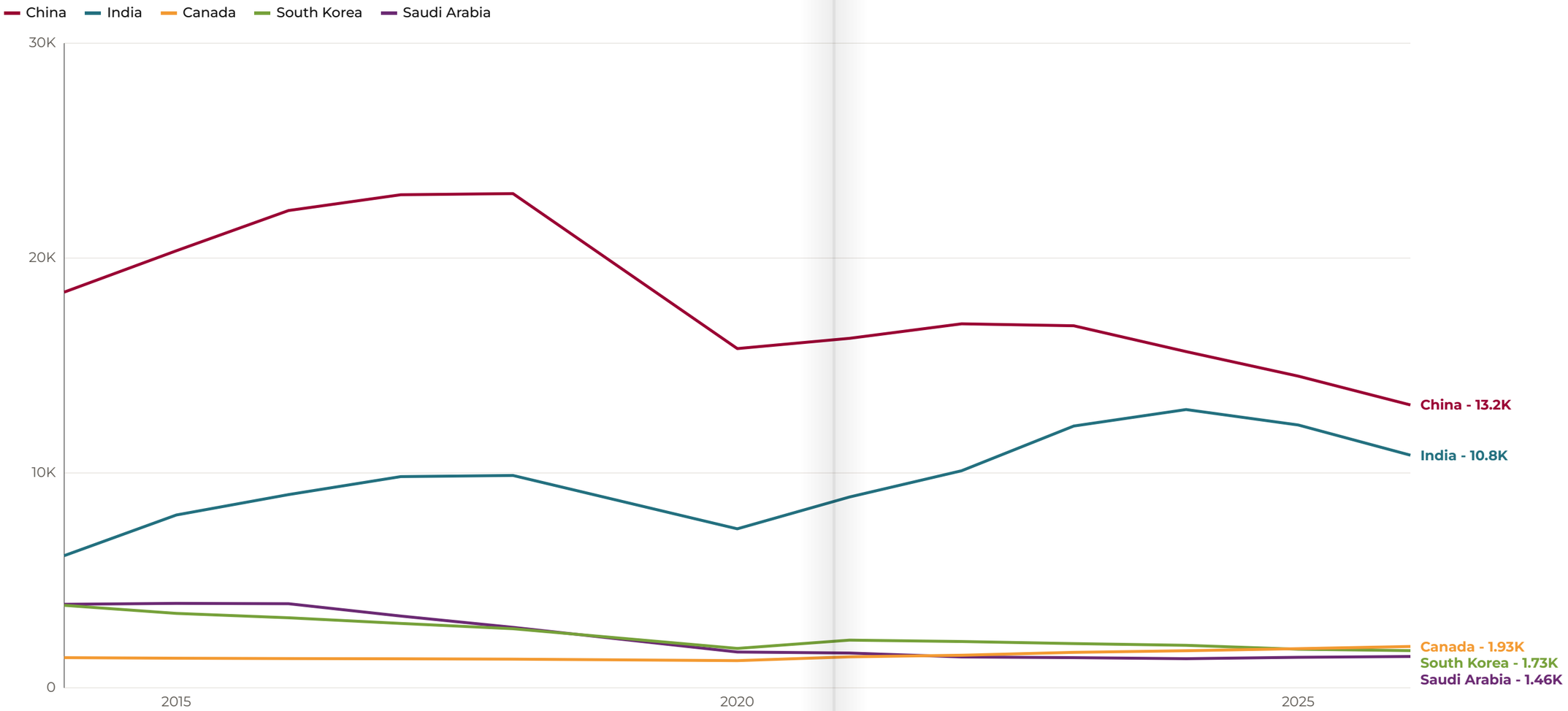
**54%**

**Int. share of STEM master's, 2025**

U.S. **52%** · **14th** of 51 · +6.9 pp since 2015

PENNSYLVANIA · STATE TALENT PROFILE

## Figure 2 Where Do Pennsylvania's International Students Come From?




Active F-1 and M-1 students by country of citizenship, the five largest in October 2026; one fall snapshot per year (the month closest to November), October 2014–October 2026. All education levels. Data from DHS, SEVIS by the Numbers.

In October 2026, Pennsylvania had 48.6K active international students in SEVIS, 7th of 51. China accounted for 27% of them and India for 22%, against 18% and 27% nationally.

Students came from 186 countries, but the two largest supplied 49%. The state's total is up 0% since 2014.

**48.6K**
**Active international students, October 2026**
**7th** of 51 · U.S. **1.19M**

**27%**
**From China**
U.S. **18%**

**64%**
**In STEM fields**
U.S. **63%**

INTERNATIONAL STUDENTS

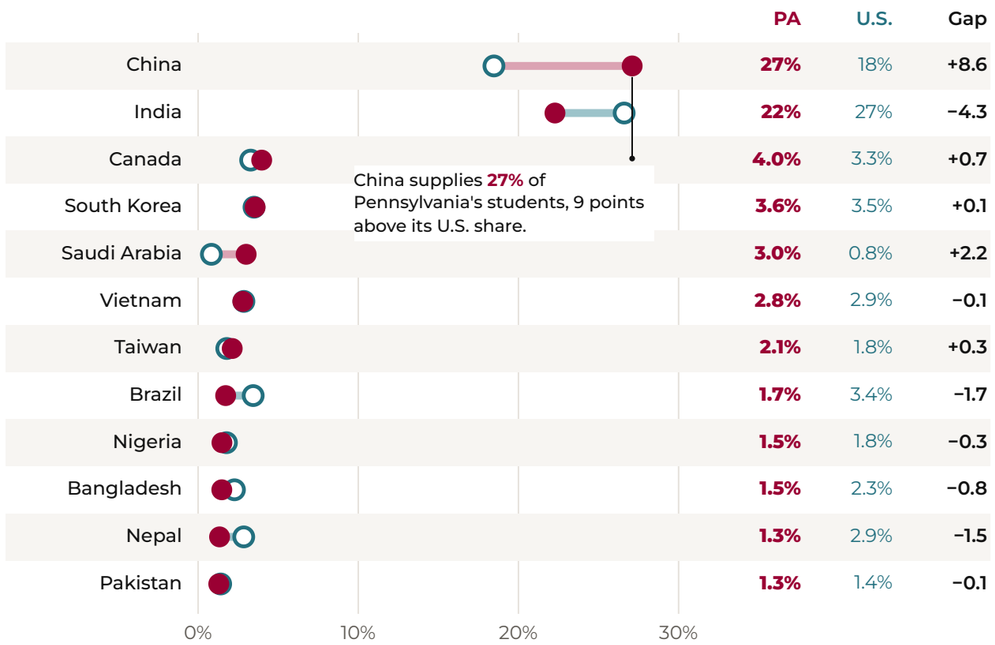
# How Does Pennsylvania's Mix of Countries Compare With the Nation's?

In October 2026, Pennsylvania had **48.6K** active F-1 and M-1 students, 7th of 51 states and **4.1%** of the U.S. total. They came from 186 countries.

China supplied **27%** of them, against 18% nationally, and India **22%**, against 27%. Canada (4.0%), South Korea (3.6%) and Saudi Arabia (3.0%) followed.

Together the two largest countries supplied **49%** of the state's international students (U.S. 45%), 15th among the states, against 54% in 2015.

## Figure 3 Is Pennsylvania's Mix of Countries Like the Nation's?




Share of active F-1 and M-1 students, all levels. Data from DHS, SEVIS by the Numbers, October 2026.





**48.6K**
Active international students, October 2026
**7th** of 51 · U.S. **1.19M**

**27%**
From China
U.S. **18%**

**22%**
From India
U.S. **27%**

**186**
Countries represented
U.S. **219**

The state's count is down 6.6% from 2025 and down 7.3% since 2015, when it stood at 52.4K. Its share of all U.S. international students is down 0.3 points since 2015, to **4.1%**.

Since 2015, students from China went from 20.3K to 13.2K; students from India went from 8.04K to 10.8K; students from Canada went from 1.38K to 1.93K.

## Table 1 International Students in Pennsylvania, 2015–2026

| | 2015 | 2025 | 2026 | Change since 2015 | U.S. 2026 | Rank |
|---|---|---|---|---|---|---|
| **ALL INTERNATIONAL STUDENTS** | | | | | | |
| Active F-1 and M-1 students | 52.4K | 52K | **48.6K** | −7.3% | 1.19M | 7th |
| Share of U.S. active F-1/M-1 students | 4.4% | 4.1% | **4.1%** | −0.3 pp | – | 7th |
| Countries of citizenship represented | 185 | 185 | **186** | +0.5% | 219 | 6th |
| Share from the two largest countries of origin | 54% | 51% | **49%** | −4.8 pp | 45% | 15th |
| **LARGEST COUNTRIES OF CITIZENSHIP (STUDENTS)** | | | | | **U.S. SHARE** | **STATE SHARE** |
| China | 20.3K | 14.5K | **13.2K** | −35% | 18% | 27% |
| India | 8.04K | 12.2K | **10.8K** | +35% | 27% | 22% |
| Canada | 1.38K | 1.82K | **1.93K** | +40% | 3.3% | 4.0% |
| South Korea | 3.46K | 1.79K | **1.73K** | −50% | 3.5% | 3.6% |
| Saudi Arabia | 3.93K | 1.42K | **1.46K** | −63% | 0.8% | 3.0% |

Active F-1 and M-1 students, one fall snapshot per year (the month closest to November; 2019 has none; the latest year is October 2026). Rank among the 50 states and DC. Data from DHS, SEVIS by the Numbers.

Figure 4

## Which Countries Supply Pennsylvania's International STEM Students?

**31.1K** of Pennsylvania's international students were in STEM fields in October 2026, **64%** of the total against 63% nationally, up 27% since 2015.

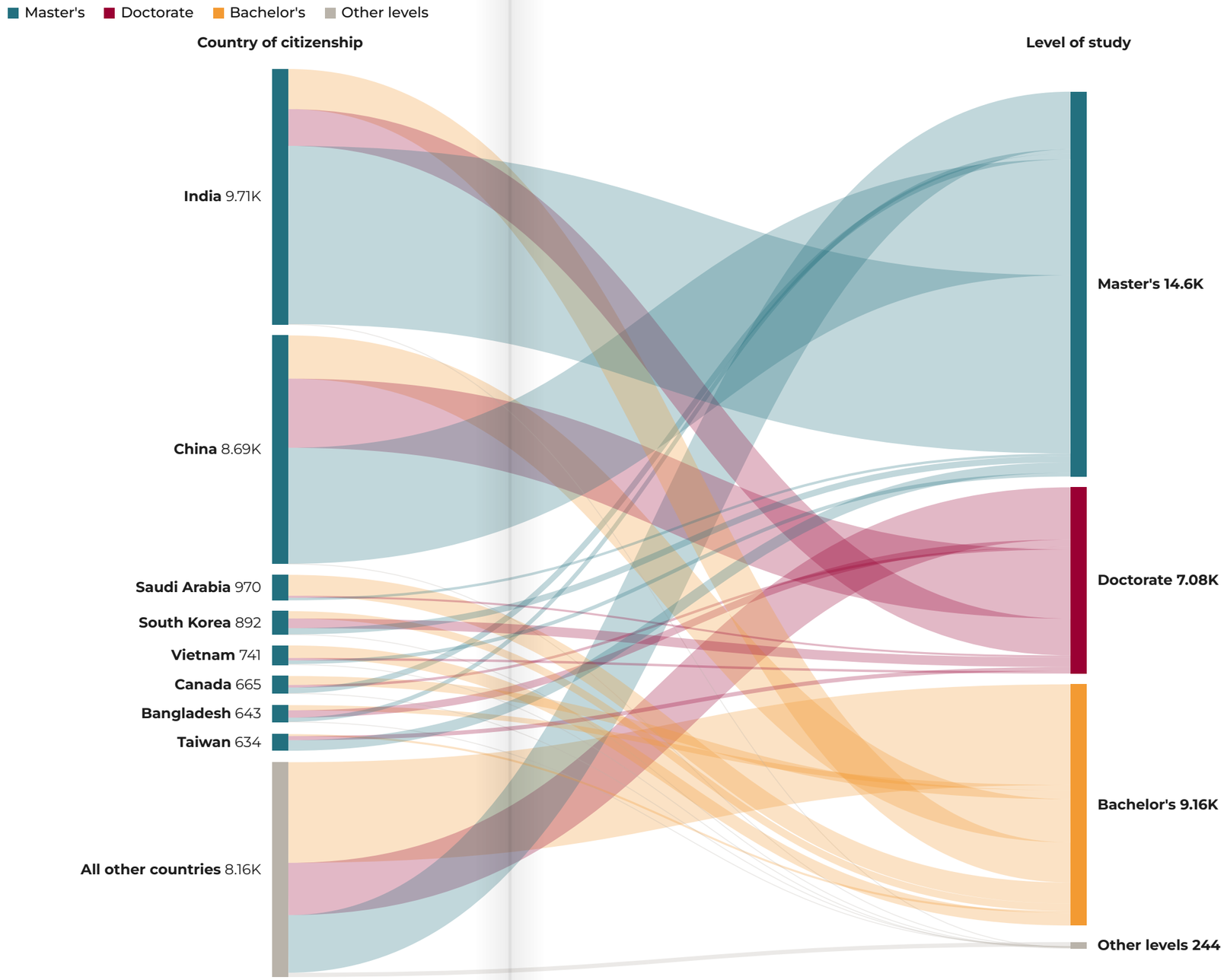
Master's students were the largest group, **14.6K**. India supplied 46% of them and China 30%.

STEM doctoral students went from 4.99K in 2015 to **7.08K**, and STEM master's students from 10.8K to 14.6K.

**31.1K**
Active STEM F-1/M-1 students, 2026
U.S. **749K** · **7th** of 51 · +27% since 2015

**64%**
Share in STEM fields
U.S. **63%**

**14.6K**
Active STEM master's students, 2026
U.S. **393K** · **7th** of 51 · +36% since 2015




Active F-1 and M-1 students in STEM fields by country of citizenship and level of study, October 2026; the eight largest countries are shown. Data from DHS, SEVIS by the Numbers, October 2026.

INTERNATIONAL DEPENDENCE

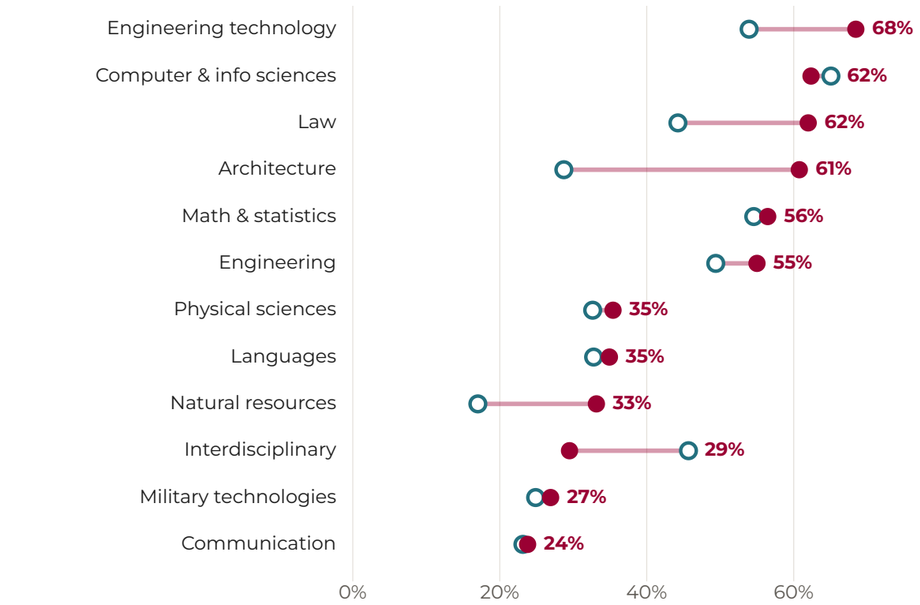
# How International Are Pennsylvania's Master's Fields?

Each row is a master's field with at least 30 graduates in Pennsylvania in 2025, ranked by the share of degrees earned by international students (U.S. nonresidents on temporary visas). The filled dot is the state's share; the open dot is the national share for the same field.

**Highest international shares.** Engineering technology 68% (U.S. 54%); computer & info sciences 62% (U.S. 65%); law 62% (U.S. 44%).

**All master's degrees.** 24% in 2025 (U.S. 22%; 12th of 51 states), up 1.7 points from 22% in 2024, and up 7.2 points since 2015 (16%).

## Figure 5 Which Master's Fields Depend Most on International Students?




International share of master's degrees by field, 2025; fields with 30+ graduates. Data from NCES, IPEDS Completions.





**24%**
Int. share of all master's, 2025
U.S. **22%** · **12th** of 51

**54%**
Int. share of STEM master's, 2025
U.S. **52%** · **14th** of 51

**62%**
Int. share of CS master's, 2025
U.S. **65%** · **27th** of 47

The table gives the international share of degrees in each STEM field at three levels: Pennsylvania first, the United States after the slash. The master's column is highlighted. A dash means the state awarded no degrees in the field at that level.

**Highest international share among STEM master's fields.** Engineering technology, 68% (U.S. 54%).

**STEM fields where international students earn a majority of master's degrees.** 5.

## Table 2 How International Is Each STEM Field?

| Field | Bachelor's | Master's | Doctorates |
|---|---|---|---|
| Computer & info sciences | 12% / 8.6% | **62% / 65%** | 65% / 56% |
| Engineering | 11% / 7.2% | **55% / 49%** | 54% / 55% |
| Engineering technology | 1.6% / 4.5% | **68% / 54%** | – / 42% |
| Math & statistics | 18% / 17% | **56% / 55%** | 50% / 53% |
| Physical sciences | 5.4% / 6.0% | **35% / 33%** | 42% / 38% |
| Biological sciences | 3.0% / 3.4% | **19% / 23%** | 21% / 23% |
| Natural resources | 2.1% / 2.0% | **33% / 17%** | – / 27% |
| Architecture | 7.1% / 7.3% | **61% / 29%** | – / 55% |

International share in Pennsylvania / U.S., 2025. Data from NCES, IPEDS Completions.

INTERNATIONAL DEPENDENCE

# How Has Pennsylvania's STEM Master's Output Grown?

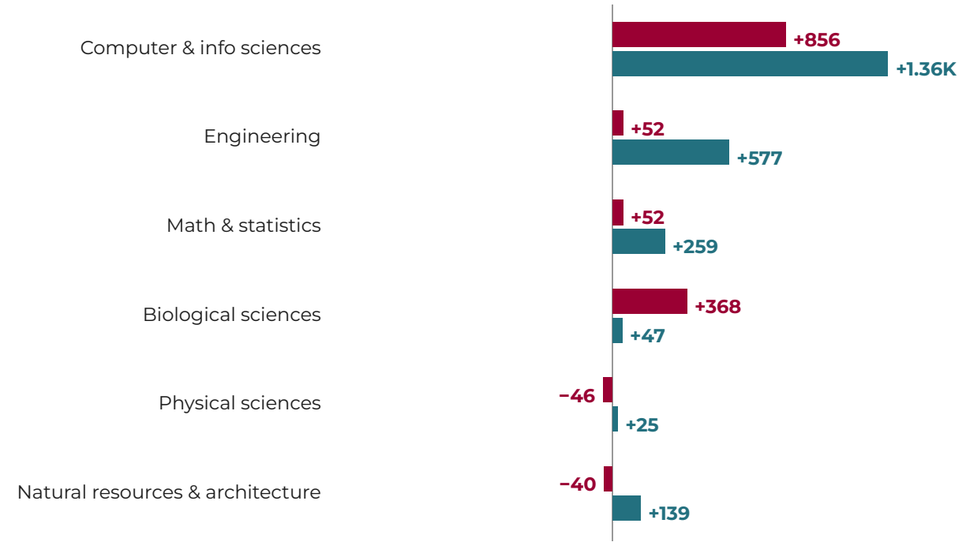
Each pair of bars is the change in the number of STEM master's degrees awarded in Pennsylvania between 2015 and 2025 in one field: to domestic students (U.S. citizens and permanent residents) and to international students. Bars to the left of the axis are declines.

**STEM master's degrees.** 10.3K in 2025, against 6.64K in 2015: +2.41K for international students and +1.24K for domestic students.

**Computer & info sciences.** +856 for domestic students and +1.36K for international students.

**Engineering.** +52 for domestic students and +577 for international students.

## Figure 6 Where Did the STEM Master's Growth Come From?




Change in STEM master's degrees by field, 2015–2025. Data from NCES, IPEDS Completions.





**5.51K**
International STEM master's, 2025
U.S. **117K** · **7th** of 51

**25%**
STEM share of master's, 2025
U.S. **23%** · **15th** of 51

**4.7%**
Share of U.S. international STEM master's, 2025
**7th** of 51

**5.4**
STEM master's per doctorate, 2025
U.S. **6.4** · **25th** of 51

The table gives the number of STEM master's degrees awarded in Pennsylvania in each field in 2015 and 2025, the change between the two years, the number awarded to international students in 2025, and international students' share of the field's degrees in each year.

**STEM share of master's degrees.** 25% in 2025 (U.S. 23%; 15th of 51 states), up 1.8 points from 23% in 2024, and up 7.0 points since 2015 (18%).

## Table 3 How Many STEM Master's Degrees Does Each Field Award?

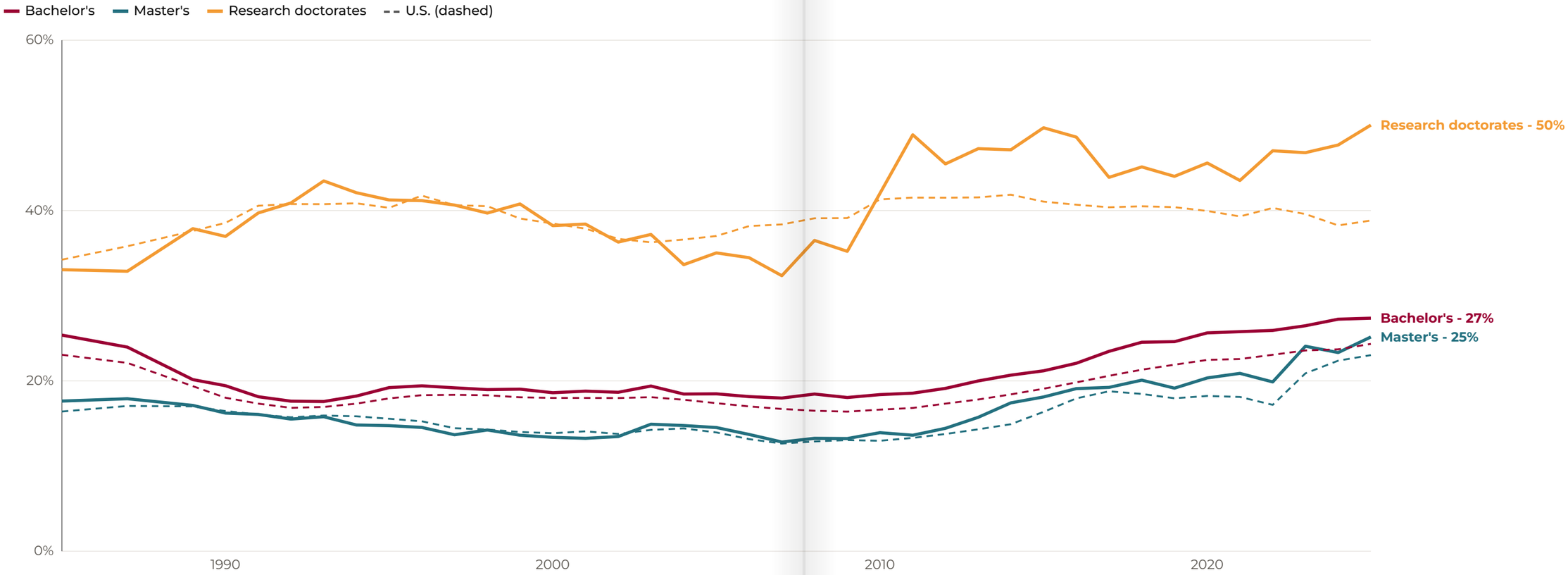
| Field | 2015 | 2025 | Change | International 2025 | Int. share 2015 | Int. share 2025 |
|---|---|---|---|---|---|---|
| Computer & info sciences | 1.83K | **4.05K** | +121% | 2.52K | 63% | 62% |
| Engineering | 2.94K | **3.57K** | +21% | 2.01K | 49% | 56% |
| Math & statistics | 355 | **666** | +88% | 376 | 33% | 56% |
| Biological sciences | 767 | **1.18K** | +54% | 225 | 23% | 19% |
| Physical sciences | 326 | **305** | −6.4% | 108 | 25% | 35% |
| Natural resources & architecture | 426 | **525** | +23% | 269 | 31% | 51% |

STEM master's degrees awarded in Pennsylvania, all students and international students, 2015 and 2025. Data from NCES, IPEDS Completions.

SECTION 2

# STEM Degrees

## Figure 7 How Much of Pennsylvania's Degree Output Is STEM?




STEM share of degrees by level; state solid, U.S. dashed. Data from NCES, IPEDS Completions.

STEM fields account for 27% of Pennsylvania's bachelor's degrees, 25% of its master's and 50% of its research doctorates (U.S. 24%, 23% and 39%).

The STEM share of master's degrees is up 7.0 points since 2015, and the state ranks 15th of 51 on it. On the STEM share of bachelor's degrees it ranks 10th of 51.

**27%**
**STEM share of bachelor's, 2025**
U.S. **24%** · **10th** of 51 · +6.2 pp since 2015

**25%**
**STEM share of master's, 2025**
U.S. **23%** · **15th** of 51 · +7.0 pp since 2015

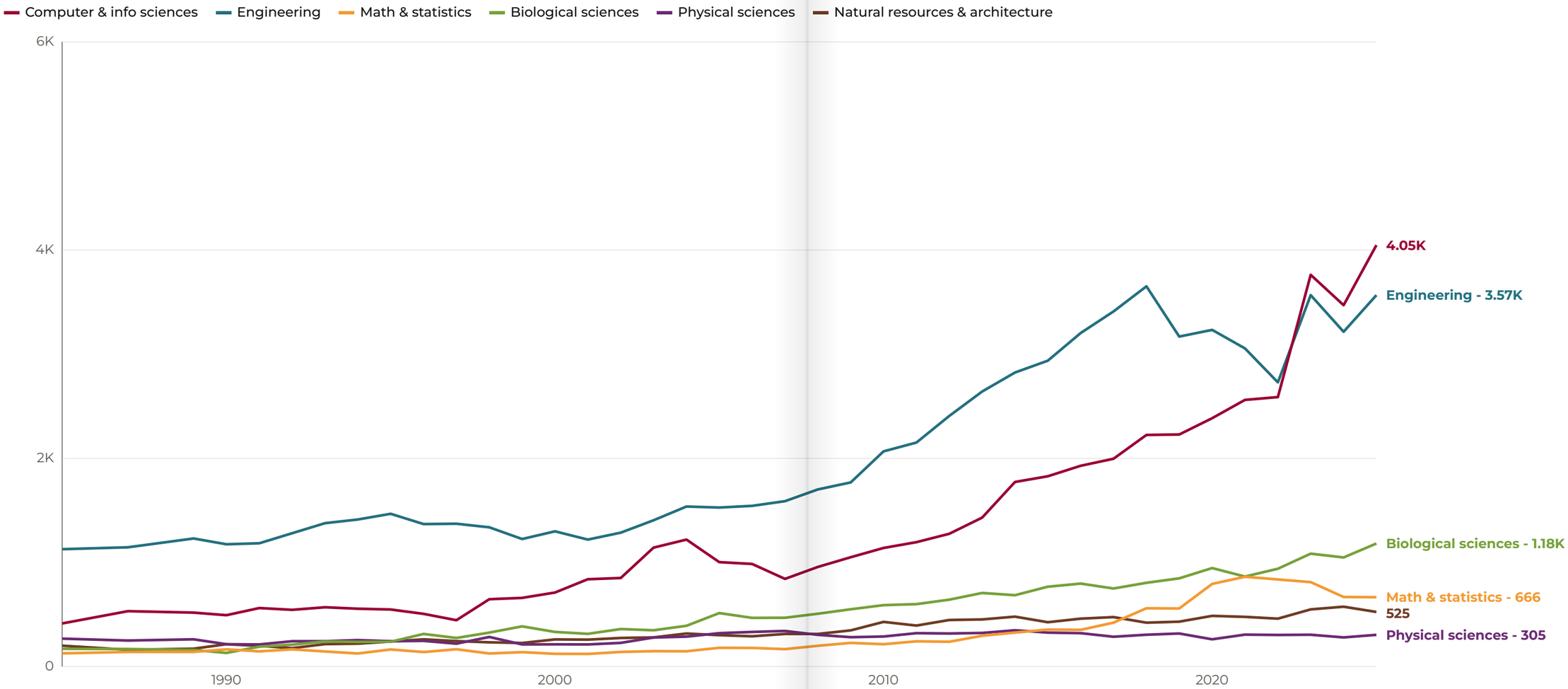
**50%**
**STEM share of research doctorates, 2025**
U.S. **39%** · **10th** of 51 · +0.3 pp since 2015



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## Figure 8 Which STEM Fields Produce Pennsylvania's Master's Degrees?




STEM master's degrees by field, all students. Data from NCES, IPEDS Completions.

Computer & info sciences produced the most STEM master's degrees in 2025, 4.05K, followed by engineering (3.57K) and biological sciences (1.18K). Master's degrees in computer & info sciences are up 121% since 2015.

International students earned 5.51K of the state's STEM master's degrees in 2025, 4.7% of the national total (7th of 51).

**5.51K**

**International STEM master's, 2025**

U.S. **117K** · **7th** of 51 · +78% since 2015

**4.7%**

**Share of U.S. international STEM master's, 2025**

U.S. **100%** · **7th** of 51 · −1.1 pp since 2015

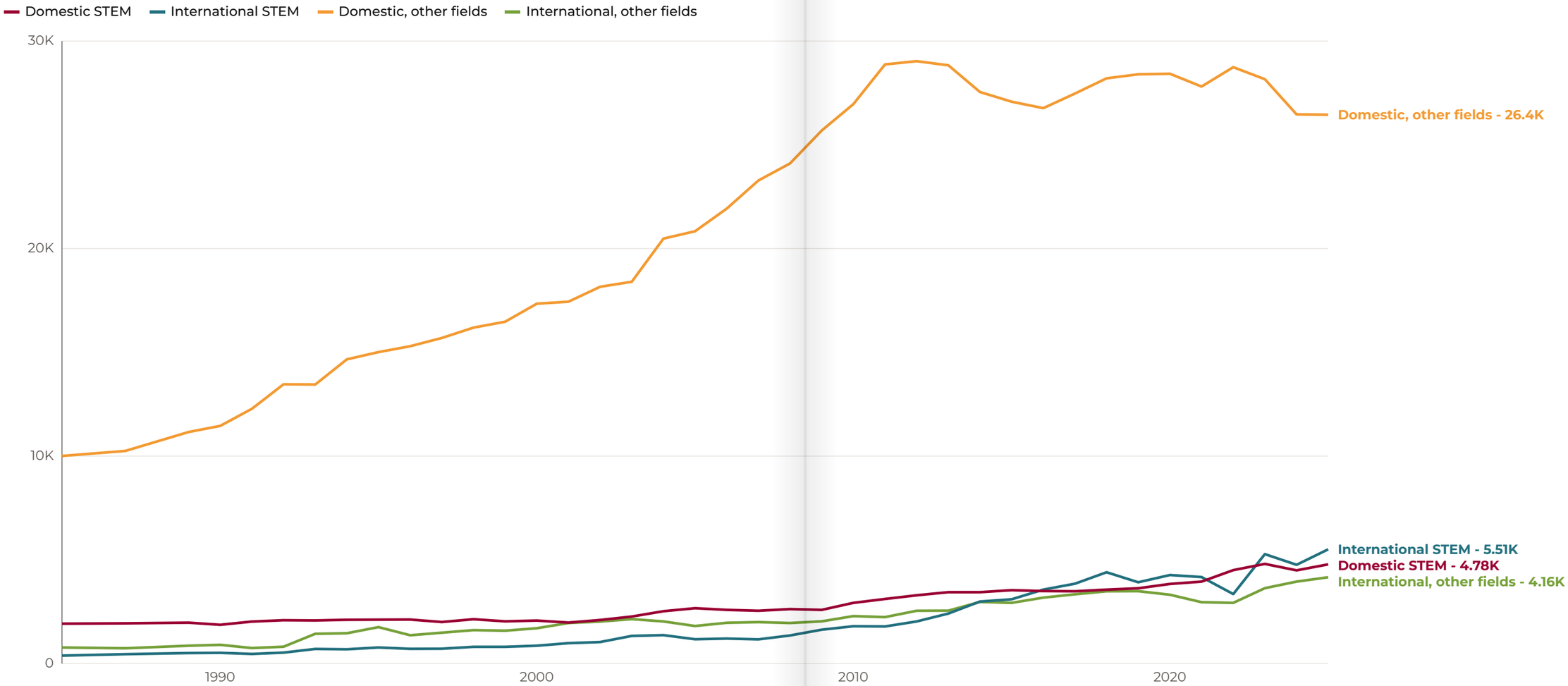
**62%**

**Int. share of CS master's, 2025**

U.S. **65%** · **27th** of 47 · −1.1 pp since 2015

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## Figure 9 Who Earns Pennsylvania's Master's Degrees?




Master's degrees by student residency and field group. Data from NCES, IPEDS Completions.

Between 2015 and 2025, 2.41K more STEM master's degrees went to international students and 1.24K more to domestic students.

International students earn 54% of Pennsylvania's STEM master's but 14% of master's in other fields (U.S. 52% and 12%).

**24%**

**Int. share of all master's, 2025**

U.S. **22%** · **12th** of 51 · +7.2 pp since 2015

**54%**

**Int. share of STEM master's, 2025**

U.S. **52%** · **14th** of 51 · +6.9 pp since 2015

**5.4**

**STEM master's per doctorate, 2025**

U.S. **6.4** · **25th** of 51 · +1.2 since 2015

SECTION 3

# Where Students Come From

## Figure 10 First-Time Students by Home and Destination, Fall 2024




Home of first-time degree-seeking undergraduates (left) and the state of the institution entered (right); the five largest sources and destinations, the rest grouped. Data from NCES, IPEDS Fall Enrollment.

73.5K residents stayed in Pennsylvania for college in fall 2024; 29.9K students arrived from other states and 3.67K from abroad, while 22.3K residents left.

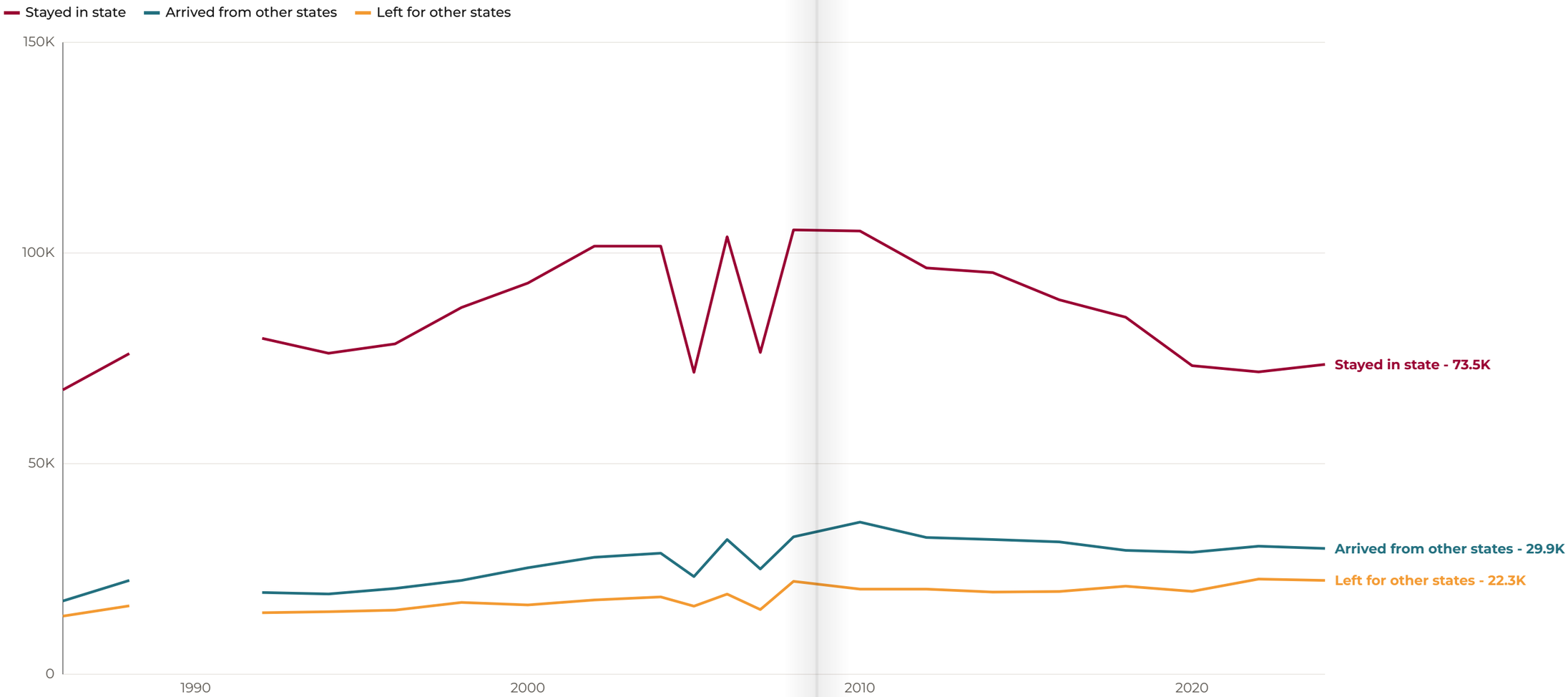
Each band runs from a student's home on the left to the state where they enrolled on the right. 68% of the state's first-time students are residents (U.S. –).

**68%**
**First-time students from in state, 2024**
· **31st** of 51 · −2.4 pp since 2016

**3.4%**
**First-time students from abroad, 2024**
U.S. **3.1%** · **16th** of 51 · −0.1 pp since 2016

**+8**
**Net migration per 100 residents, 2024**
· **21st** of 51 · −2.9 since 2016

## Figure 11 How Have the Flows Changed?




First-time degree-seeking undergraduates by flow; even years before 2001. Data from NCES, IPEDS Fall Enrollment.

Since 2015, the number of residents staying in state for college is down 17%, arrivals from other states are down 5%, and residents leaving are up 13%.



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Net migration was +8 per 100 resident first-time students (21st of 51). The top destination for residents who left was New York (2.57K).

Figure 12

## Where Do Pennsylvania's Newcomers Come From?

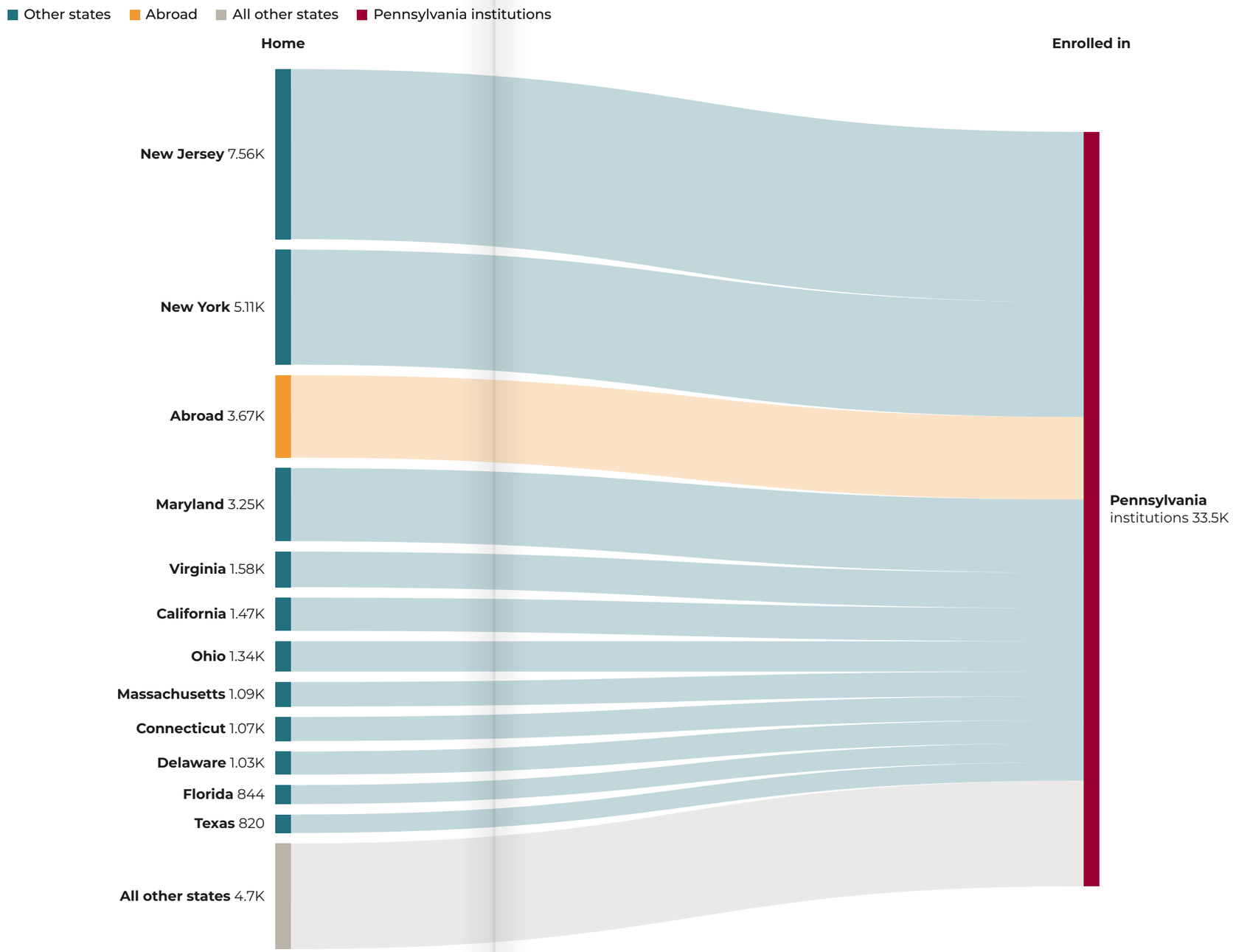
The bands show where first-time degree-seeking undergraduates who enrolled at Pennsylvania's institutions in fall 2024 but did not live in the state came from: the 12 largest home states and students from abroad, with all other states grouped. Band widths are proportional to the number of students. Home is the state where the student lived when admitted.

**Students arriving from other states.** 29.9K in fall 2024, down 1.8% from 30.4K in fall 2022, and down 5.0% since fall 2016 (31.4K).

**33.5K**
First-time students from other states and abroad, fall 2024
Abroad **3.67K**

**3.4%**
First-time students from abroad, 2024
U.S. **3.1%** · **16th** of 51 · −0.1 pp since 2016

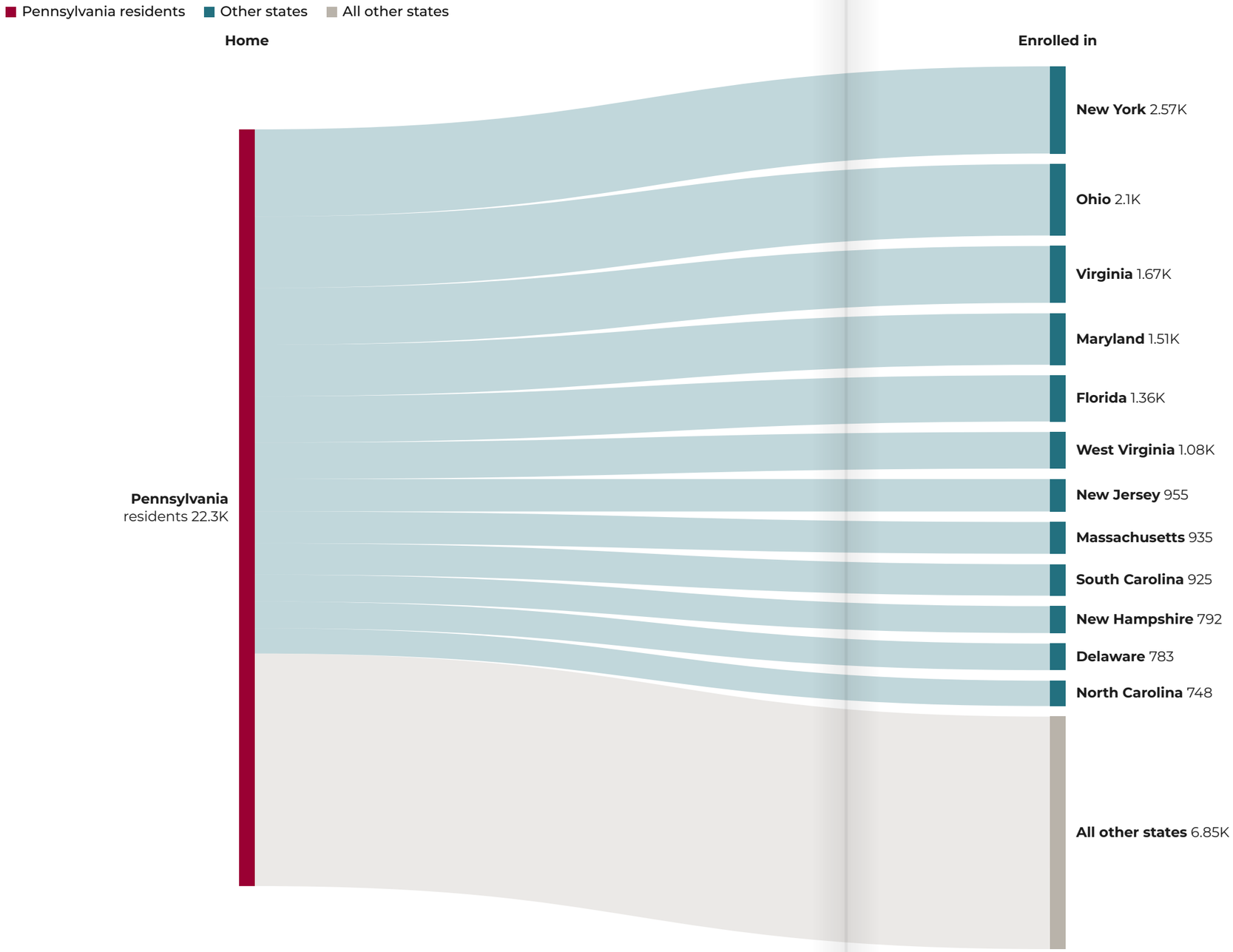
**68%**
First-time students from in state, 2024
**31st** of 51 · −2.4 pp since 2016




First-time degree-seeking undergraduates, fall 2024; the 12 largest flows. Data from NCES, IPEDS Fall Enrollment.



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First-time degree-seeking undergraduates, fall 2024; the 12 largest flows. Data from NCES, IPEDS Fall Enrollment.





Figure 13

## Where Do Pennsylvania's Residents Go When They Leave?

The bands show where Pennsylvania residents who left the state to start college in fall 2024 enrolled: the 12 largest destination states, with all other states grouped. Band widths are proportional to the number of students. Residents who enrolled in the state are not shown.

**Residents leaving for other states.** 22.3K in fall 2024, down 1.5% from 22.6K in fall 2022, and up 13% since fall 2016 (19.6K).

**Residents staying in the state.** 73.5K in fall 2024, up 2.5% from 71.8K in fall 2022, and down 17% since fall 2016 (88.9K).

**22.3K**
Residents who started college in another state, fall 2024
Arrivals **33.5K**

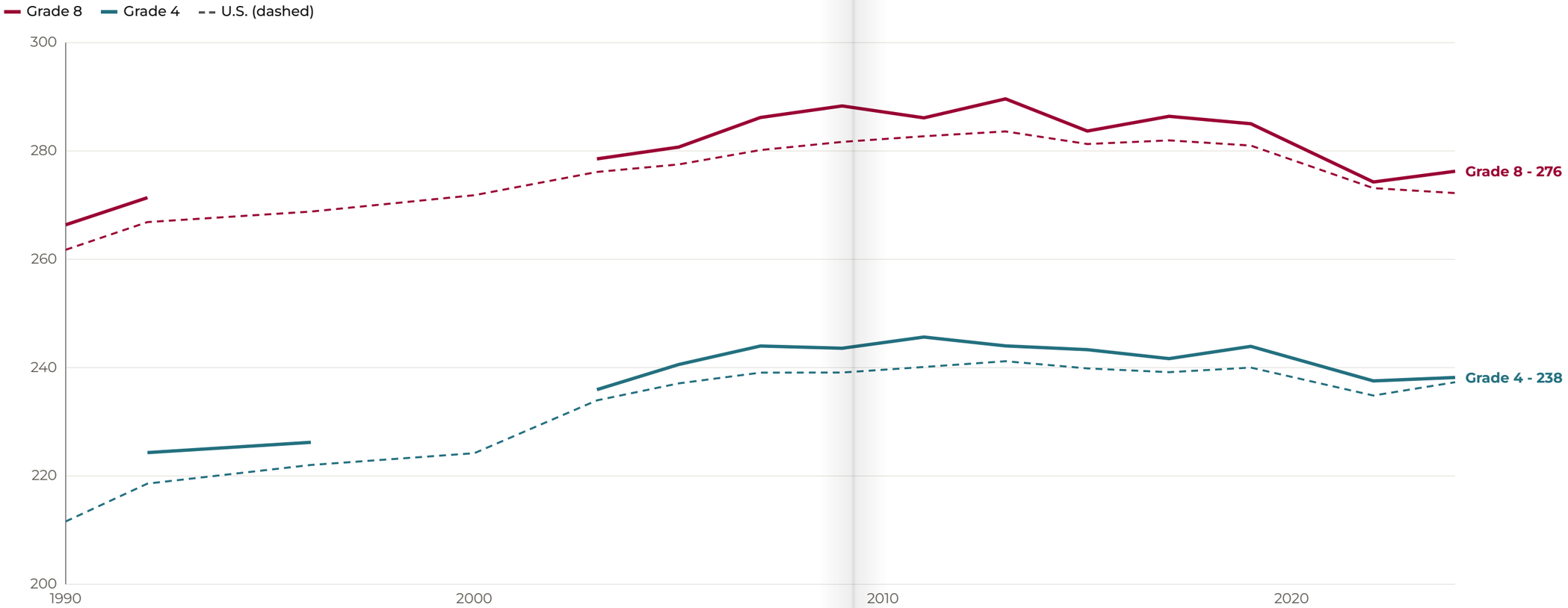
**+8**
Net migration per 100 residents, 2024
**21st** of 51 · −2.9 since 2016

**2.57K**
Went to New York
The largest destination

SECTION 4

# K-12 Preparation

## Figure 14 How Well Prepared in Mathematics Are Pennsylvania's Students?




Average NAEP mathematics score, public schools, 0–500 scale; state solid, U.S. public schools dashed. Data from NCES, National Assessment of Educational Progress.

The talent pipeline starts in school. Pennsylvania's eighth graders averaged 276 in mathematics on the 2024 NAEP, against 272 for U.S. public schools, 18th of 51 among the states.

31% scored at or above the Proficient level (U.S. 27%). Differences of a point or two are often within the margin of error.

**276**

**Grade 8 math score (NAEP), 2024**

U.S. **272** · **18th** of 51 · −7.4 since 2015

**31%**

**Grade 8 math, at or above Proficient, 2024**

U.S. **27%** · **15th** of 51 · −4.6 pp since 2015

**259**

**Grade 8 reading score (NAEP), 2024**

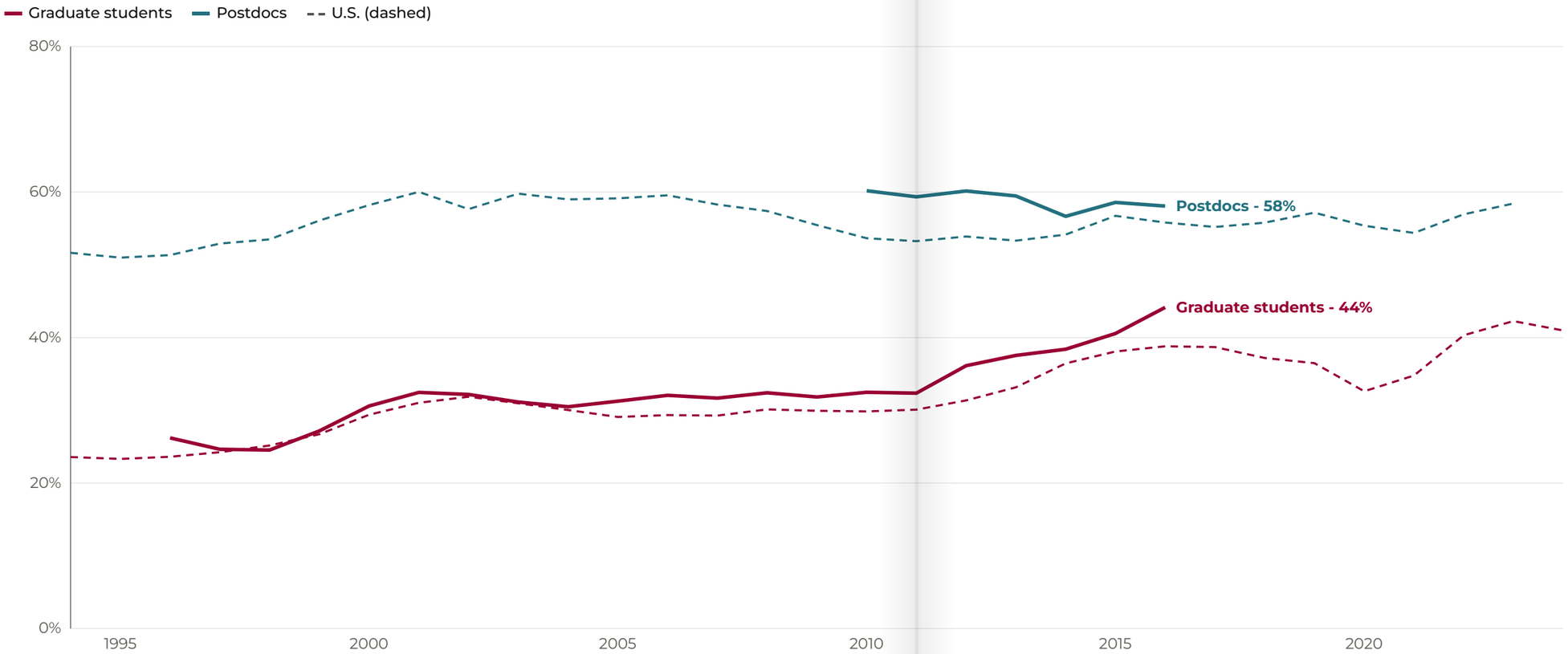
U.S. **257** · **19th** of 51 · −9.6 since 2015

PENNSYLVANIA · STATE TALENT PROFILE

SECTION 5

# Graduate Research

## Figure 15 How Many of Pennsylvania's Science and Engineering Graduate Students Hold Temporary Visas?




Temporary-visa share of science and engineering graduate students and postdoctoral appointees, by the institution's state; state solid, U.S. dashed. NSF stopped publishing state tables after 2016. Data from NSF NCSES, Survey of Graduate Students and Postdoctorates.

In 2016, the last year NSF published state tables, 44% of Pennsylvania's science and engineering graduate students held temporary visas (U.S. 39%; 10th of 51).

Among postdoctoral researchers, the share was 58% (U.S. 56%). Postdocs are the stage of research training most dependent on international talent, and IPEDS does not count them.

The state's institutions awarded 2.94K research doctorates in 2025, 5.0% of the U.S. total (Data from NSF NCSES, Survey of Earned Doctorates; NSF publishes doctorates by citizenship only for the nation).

**44%**
**S&E grad students on temporary visas, 2016**
U.S. **39%** · **10th** of 51 · +3.6 pp since 2015

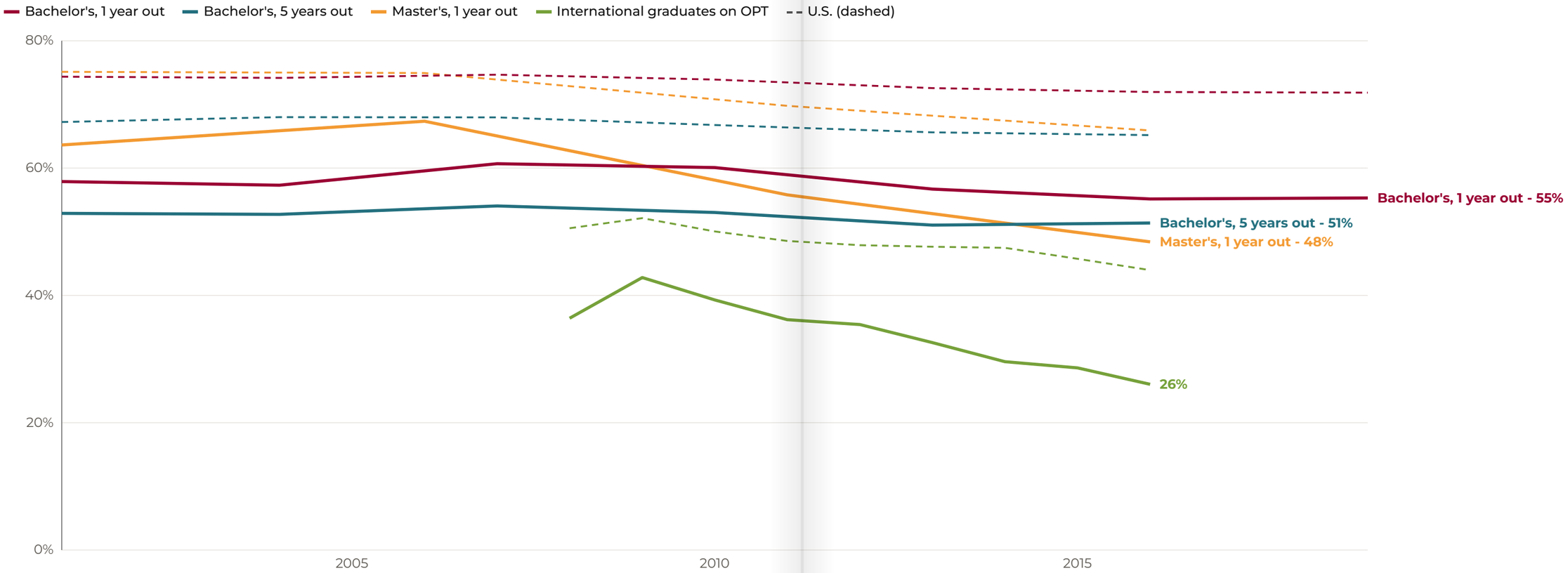
**58%**
**S&E postdocs on temporary visas, 2016**
U.S. **56%** · **18th** of 51 · −0.5 pp since 2015

**2.94K**
**Research doctorates awarded (NSF SED), 2025**
U.S. **59K** · **5th** of 51 · +12% since 2015

SECTION 6

# Do Graduates Stay?

## Figure 16 How Many of Pennsylvania's Graduates Go On to Work in the State?




Share of employed graduates working in the state of their institution, by graduating cohort (Data from U.S. Census Bureau, Post-Secondary Employment Outcomes; participating institutions only); international graduates' OPT jobs in the state where they studied (Data from ICE, SEVIS FOIA releases; shares only, the file is truncated). State solid, U.S. or participating states dashed.

Of Pennsylvania's bachelor's graduates who were working a year after graduating, 55% worked in the state (participating states 72%). Five years out the share was 51%.

Among international graduates on Optional Practical Training in 2016, 26% worked in the state where they studied (U.S. 44%). Through migration between states, the state lost 2.6 bachelor's-plus adults per 1,000 such residents in 2025 (Data from U.S. Census Bureau, American Community Survey).

**55%**

**Bachelor's grads working in-state, 1 year out, 2019**

U.S. **72%** · **27th** of 33 · +0.2 pp since 2016

**26%**

**OPT graduates working in the state, 2016**

U.S. **44%** · **38th** of 51 · −2.6 pp since 2015

**−2.6**

**Net migration, bachelor's-plus, per 1,000, 2025**

· **34th** of 51 · +2.3 since 2015

SECTION 7

# Research Funding

## Figure 17 How Much Research Money Flows Into Pennsylvania?




Millions of nominal dollars, fiscal years. University R&D: Data from NSF NCSES, Higher Education Research and Development Survey (FY2003-09 predecessor survey). Federal R&D obligations by state of performance: Data from NSF NCSES, Survey of Federal Funds for R&D (coverage widens in FY2021). Grants from NSF, NIH, NASA and DOE Science plus R&D contracts: Data from USAspending.gov.

Pennsylvania's universities spent $6.45B on research and development in FY2024, 5th of 51 among the states; 56% of it was federally funded (U.S. 55%).

Federal agencies obligated $5.15B for R&D performed in the state in FY2024. Research funding drives demand for graduate students and postdocs, the stage of training most dependent on international talent.

**$6.45B**

**University R&D spending, 2024**

U.S. **$118B** · **5th** of 51 · +92% since 2015

**56%**

**Federal share of university R&D, 2024**

U.S. **55%** · **17th** of 51 · −8.9 pp since 2015

**$5.15B**

**Federal R&D obligations, 2024**

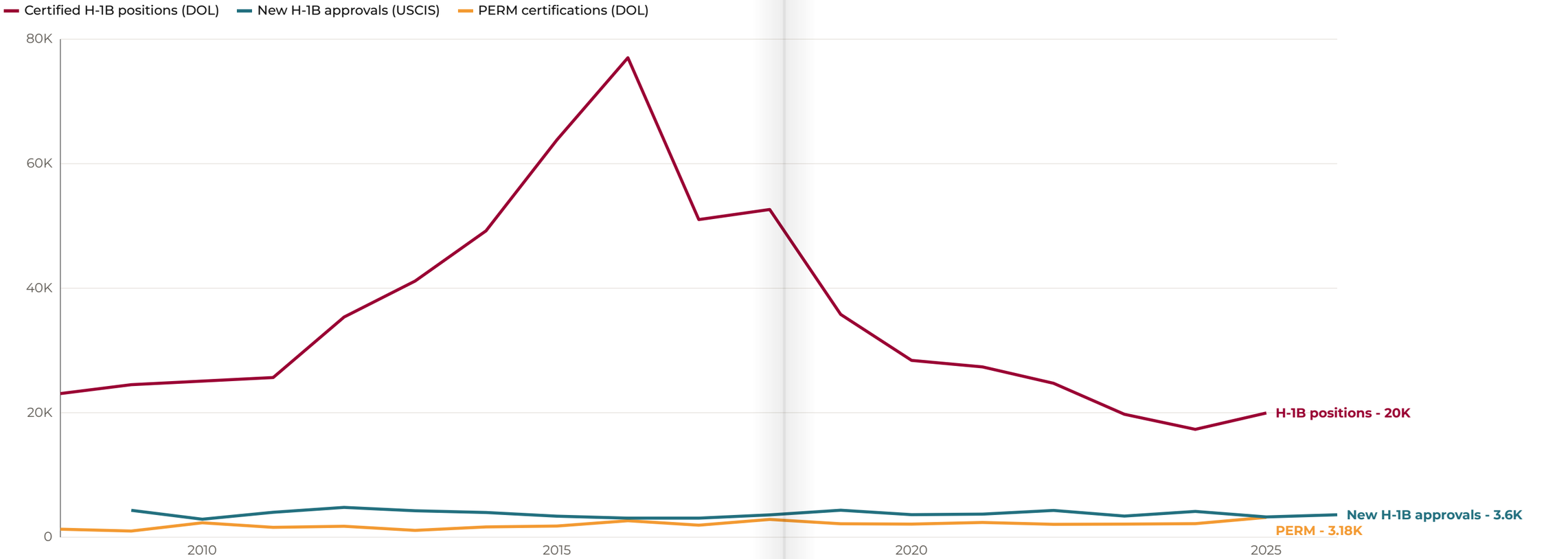
U.S. **$193B** · **12th** of 51 · +39% since 2015

PENNSYLVANIA · STATE TALENT PROFILE

SECTION 8

# H-1B Jobs and Green Cards

## Figure 18 How Many Skilled-Visa Jobs and Green-Card Certifications Does Pennsylvania Generate?




Certified H-1B positions on labor condition applications by worksite and certified PERM green-card applications, by fiscal year (Data from DOL, Office of Foreign Labor Certification); new H-1B petition approvals by the employer's address (Data from USCIS, H-1B Employer Data Hub).

Employers were certified for 20K H-1B positions at Pennsylvania worksites in FY2025, 12th of 51 among the states. A labor certification comes before the petition, so this overstates hires; USCIS approved 3.6K new H-1B petitions from employers in the state in FY2026.

DOL certified 3.18K PERM applications, the employer's step toward an employment-based green card, for jobs in the state in FY2025. In FY2024, 9.9% were for workers still on F-1 student visas (U.S. 8.7%): the most direct link from classroom to permanent residence.

**20K**

**Certified H-1B positions (worksite), 2025**

U.S. **914K** · **12th** of 51 · −69% since 2015

**3.18K**

**PERM green-card certifications, 2025**

U.S. **138K** · **13th** of 51 · +77% since 2015

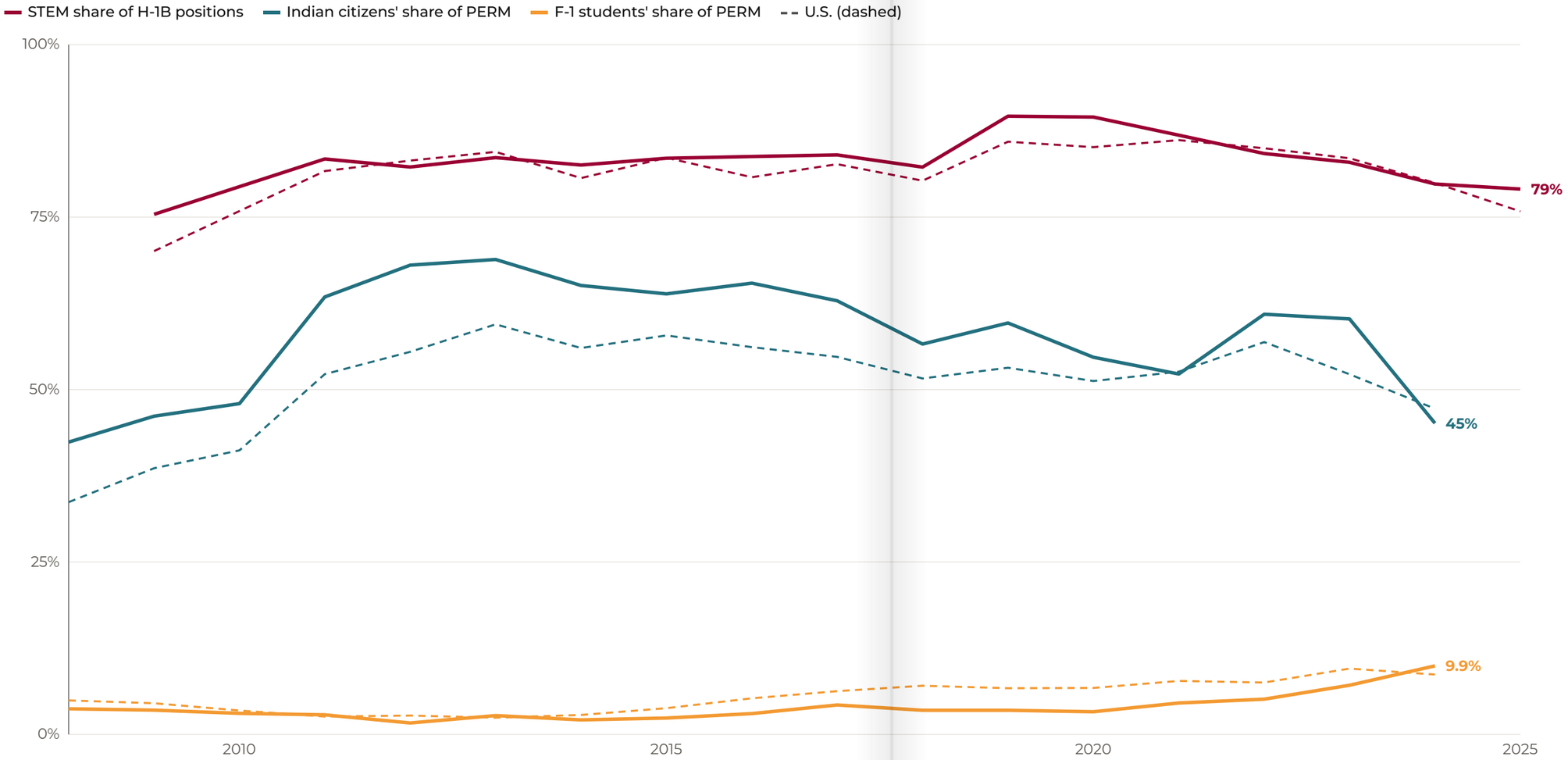
**9.9%**

**PERM workers on F-1 student visas, 2024**

U.S. **8.7%** · **24th** of 51 · +7.6 pp since 2015

PENNSYLVANIA · STATE TALENT PROFILE

## Figure 19 Who Holds Pennsylvania's Skilled-Visa Jobs?




State solid, U.S. dashed. PERM citizenship and class of admission are not on the form DOL uses from FY2025. Data from DOL, Office of Foreign Labor Certification.

79% of the state's certified H-1B positions were in STEM occupations in FY2025 (U.S. 76%).

Citizens of India held 45% of the state's PERM certifications in FY2024 (U.S. 47%). Under per-country caps they face the longest waits for a green card, so this share measures the state's exposure to the backlog.

**79%**

**STEM share of H-1B positions, 2025**

U.S. **76%** · **11th** of 51 · −4.5 pp since 2015

**45%**

**PERM certifications for Indian citizens, 2024**

U.S. **47%** · **19th** of 51 · −18.7 pp since 2015

**$103K**

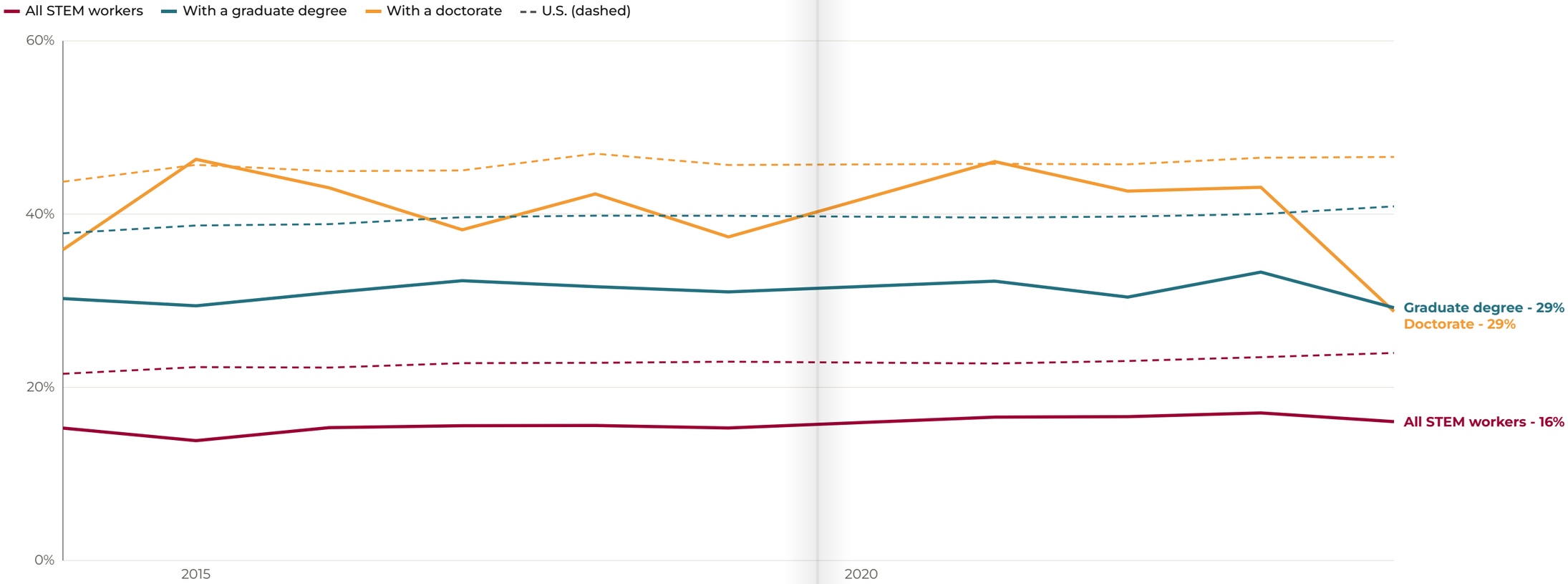
**Median offered H-1B wage (nominal), 2025**

U.S. **$119K** · **27th** of 51 · +49% since 2015

SECTION 9

# STEM Workforce

## Figure 20 How Much of Pennsylvania's STEM Workforce Was Born Abroad?




Foreign-born share of employed workers in STEM occupations, by state of residence; state solid, U.S. dashed. Shares resting on fewer than 100 survey records are left out. No 2020 (experimental data). Data from U.S. Census Bureau, American Community Survey.

In 2024, 16% of the people working in STEM occupations in Pennsylvania were born abroad (U.S. 24%; 21st of 51).

The share rises with education: 29% of STEM workers with a graduate degree were foreign-born (U.S. 41%). 7.5% of all STEM workers were not U.S. citizens.

**16%**

**Foreign-born share of STEM workers, 2024**

U.S. **24%** · **21st** of 51 · +2.2 pp since 2015

**29%**

**Foreign-born, STEM workers with grad degrees, 2024**

U.S. **41%** · **23rd** of 44 · −0.2 pp since 2015

**7.5%**

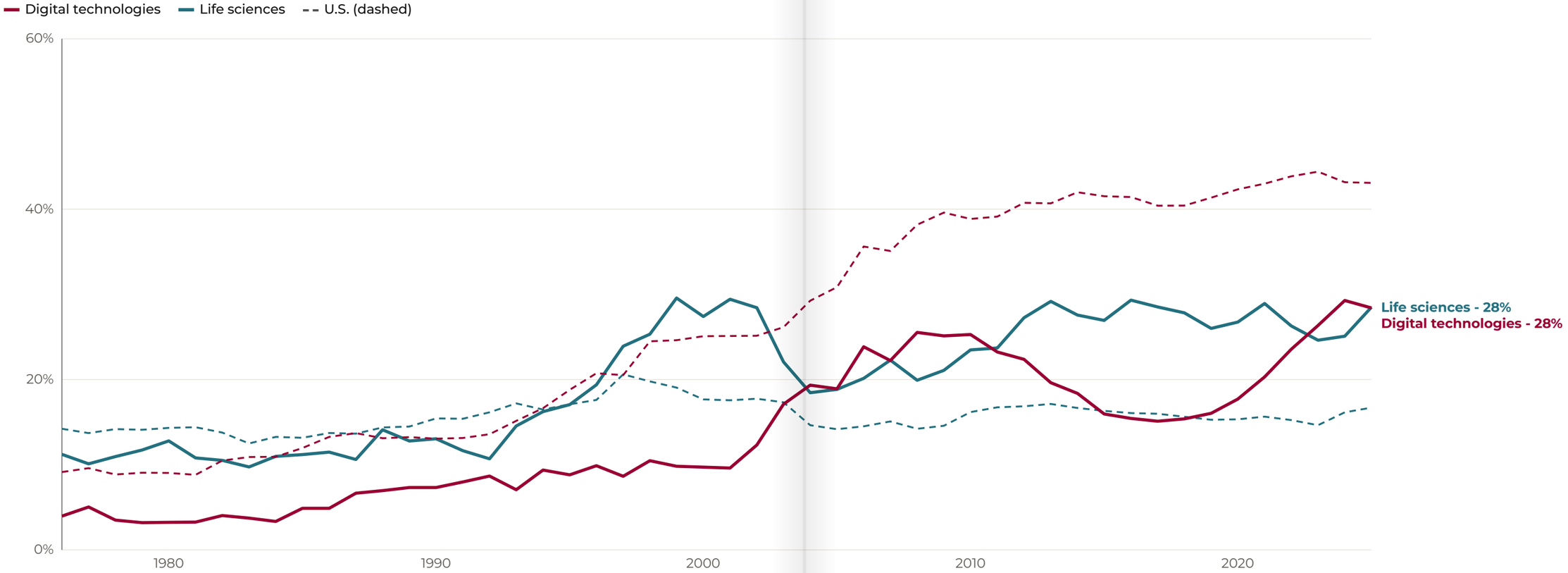
**Noncitizen share of STEM workers, 2024**

U.S. **11%** · **22nd** of 51 · +0.3 pp since 2015

SECTION 10

# Patents

## Figure 21 What Kinds of Technology Does Pennsylvania Patent?




Share of patents granted to owners in the state, by WIPO technology field of the first classification; state solid, U.S. dashed. Fractional counts by the owner's (assignee's) location, not where inventors live. Data from USPTO, PatentsView.

Organizations and people in Pennsylvania were granted 3.2K U.S. patents in 2025, 2.4% of all patents with a U.S. owner (12th of 51).

28% were in digital technologies (U.S. 43%) and 28% in the life sciences (U.S. 17%).

**3.2K**

**Patents granted to owners in the state, 2025**

U.S. **135K** · **12th** of 51 · +16% since 2015

**28%**

**Patents in digital technologies, 2025**

U.S. **43%** · **17th** of 51 · +12.5 pp since 2015

**28%**

**Patents in life sciences, 2025**

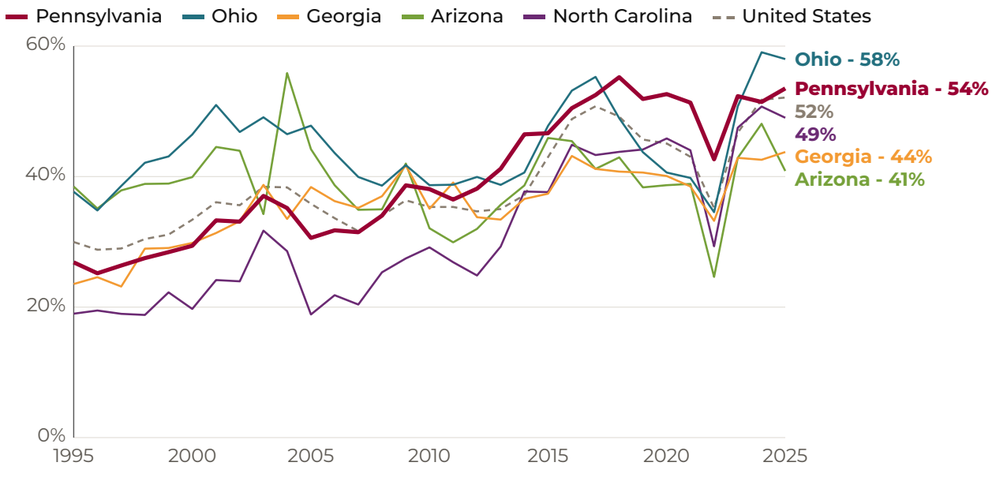
U.S. **17%** · **10th** of 51 · +1.6 pp since 2015

SECTION 11

# Competitors

The four states closest in size are Ohio, Georgia, Arizona, North Carolina. Pennsylvania ranks 2nd of the five on the international share of STEM master's; Ohio is the most dependent at 58%.

**Figure 22 How Dependent Are Pennsylvania and Its Peers on International STEM Master's?**




Data from NCES, IPEDS Completions.

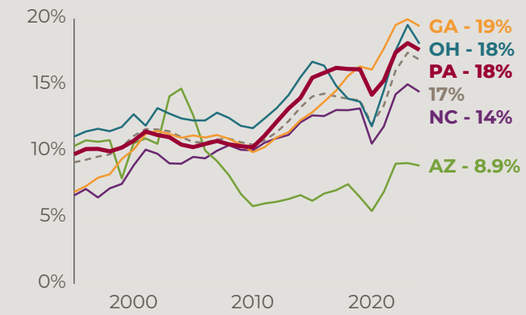
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**Table 4 How Does Pennsylvania Compare With States of Similar Size?**

| Measure | PA | OH | GA | AZ | NC | U.S. |
|---|---|---|---|---|---|---|
| Students enrolled | **637K** | 611K | 594K | 684K | 582K | 19.7M |
| Graduate share of students | **23%** | 14% | 18% | 18% | 14% | 17% |
| STEM share of bachelor's | **27%** | 24% | 27% | 17% | 26% | 24% |
| STEM share of master's | **25%** | 27% | 28% | 15% | 22% | 23% |
| International STEM master's | **5.51K** | 3.84K | 3.78K | 2.58K | 2.42K | 117K |
| Share of U.S. international STEM master's | **4.7%** | 3.3% | 3.2% | 2.2% | 2.1% | 100% |
| Int. share of STEM master's | **54%** | 58% | 44% | 41% | 49% | 52% |
| Int. share of STEM research doctorates | **45%** | 45% | 44% | 42% | 39% | 43% |
| Int. share of grad students | **18%** | 18% | 19% | 8.9% | 14% | 17% |
| Int. share of undergraduate enrollment | **3.9%** | 3.3% | 2.6% | 1.8% | 2.1% | 3.0% |
| Six-year graduation rate | **68%** | 62% | 57% | 58% | 66% | 65% |
| In-state tuition, public 4-year | **$17K** | $11K | $7.14K | $10K | $7.54K | $9.71K |
| Appropriations per student | **$7.14K** | $7.76K | $10.8K | $9.12K | $15.6K | $11.4K |
| Average faculty salary | **$109K** | $93.3K | $95.3K | $102K | $95.1K | $106K |
| First-time students from in state | **68%** | 76% | 82% | 61% | 79% | – |
| Net migration per 100 residents | **+8** | +5 | −6 | +37 | +4 | – |

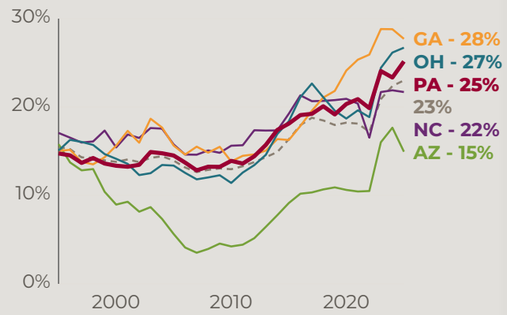
Latest year for each measure. Data from NCES, IPEDS.

**Figure 23 Int. share of grad students**



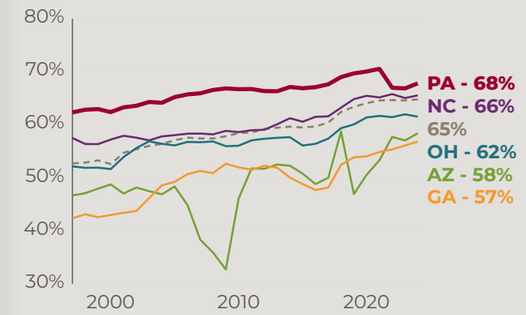

Data from NCES, IPEDS.

**Figure 24 STEM share of master's**



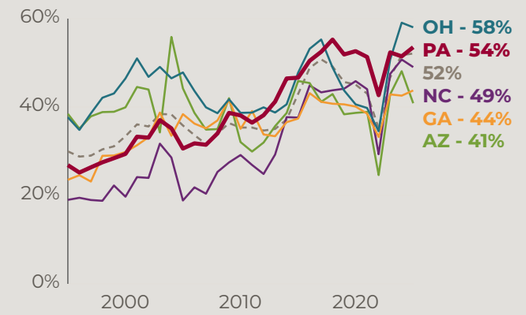

Data from NCES, IPEDS.

**Figure 25 Six-year graduation rate**




Data from NCES, IPEDS.

**Figure 26 Int. share of STEM master's**

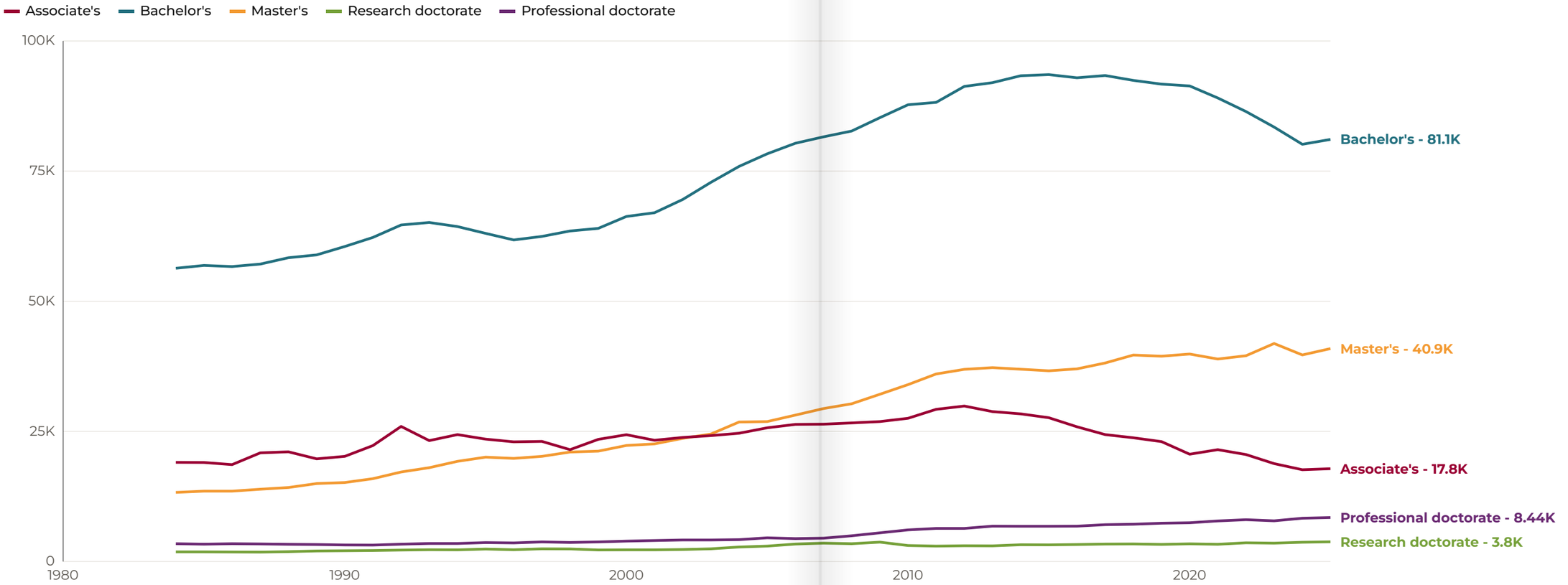



Data from NCES, IPEDS.

SECTION 12

# Fields of Study

## Figure 27 How Many Degrees Does Pennsylvania Award?

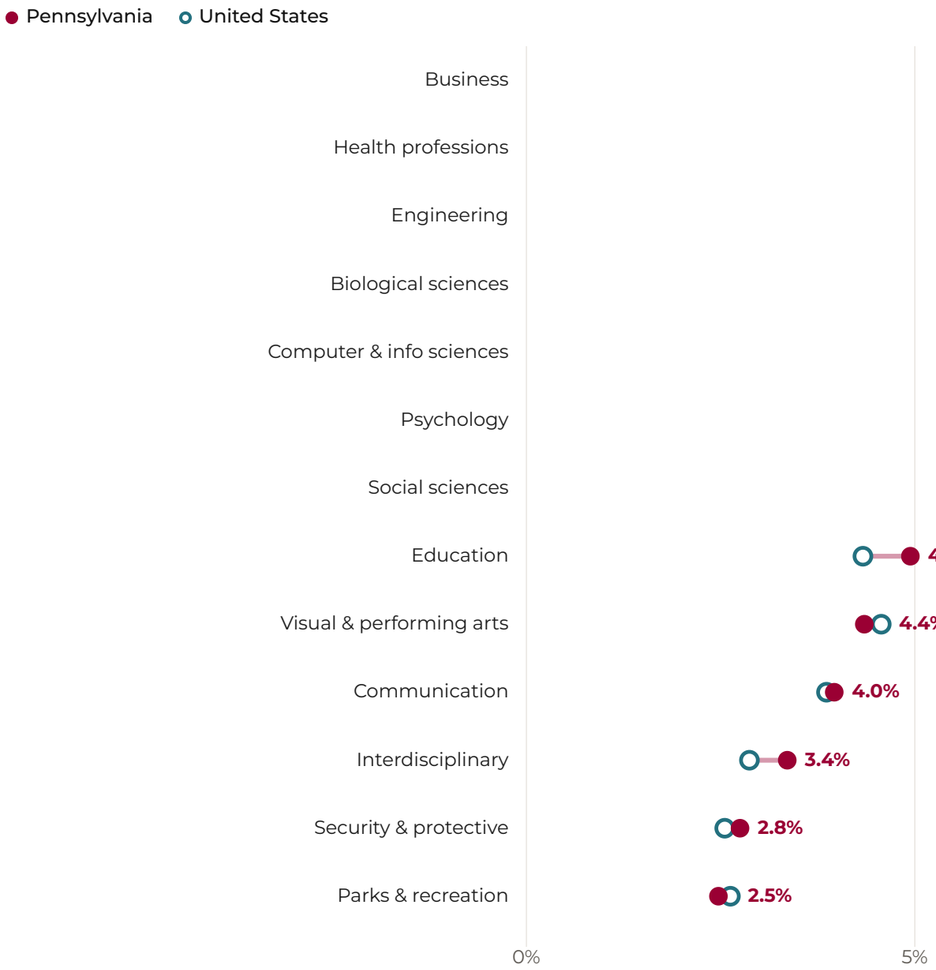


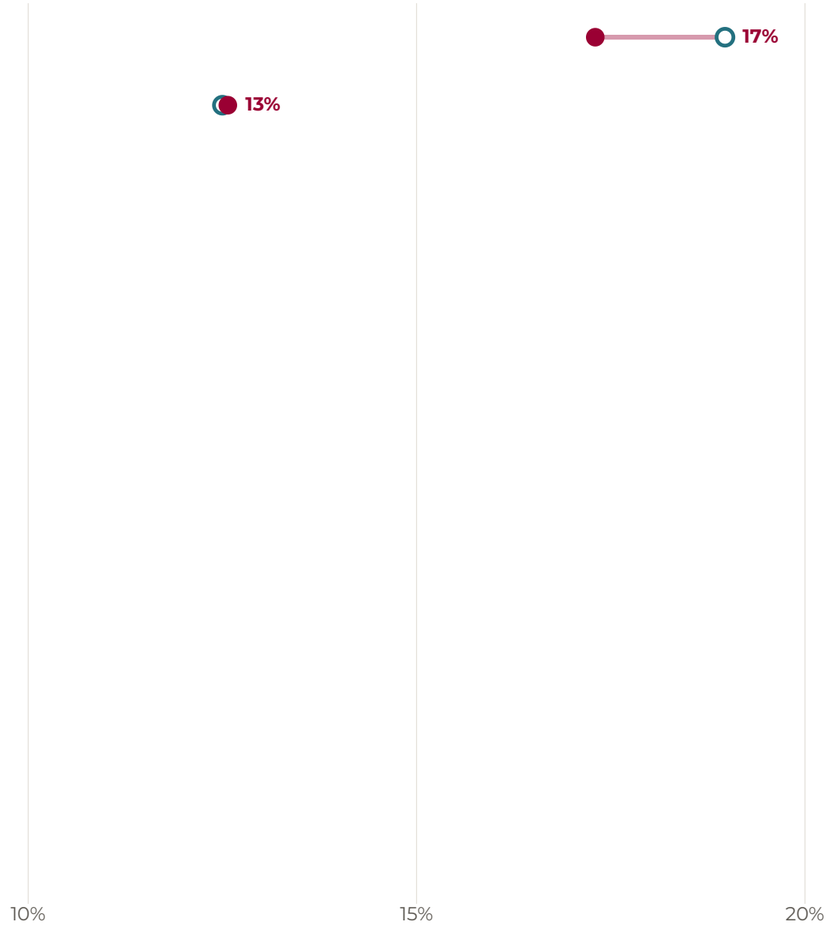

Degrees awarded by level, first majors. Data from NCES, IPEDS Completions.

In 2025, Pennsylvania's institutions awarded 17.8K associate's degrees, 81.1K bachelor's degrees, 40.9K master's degrees, 3.8K research doctorate degrees, 8.44K professional doctorate degrees.

Since 2015: associate's down 35%; bachelor's down 13%; master's up 12%; research doctorate up 19%; professional doctorate up 25%. The state awards 16.5 bachelor's degrees for every 100 undergraduates (U.S. 12.3).

## Figure 28 What Do Pennsylvania's Bachelor's Graduates Study?




Share of bachelor's degrees by field, 2025. Data from NCES, IPEDS.

Pennsylvania's largest bachelor's fields are business (17%; U.S. 19%), health professions (13%; U.S. 13%), engineering (7.3%; U.S. 5.9%).

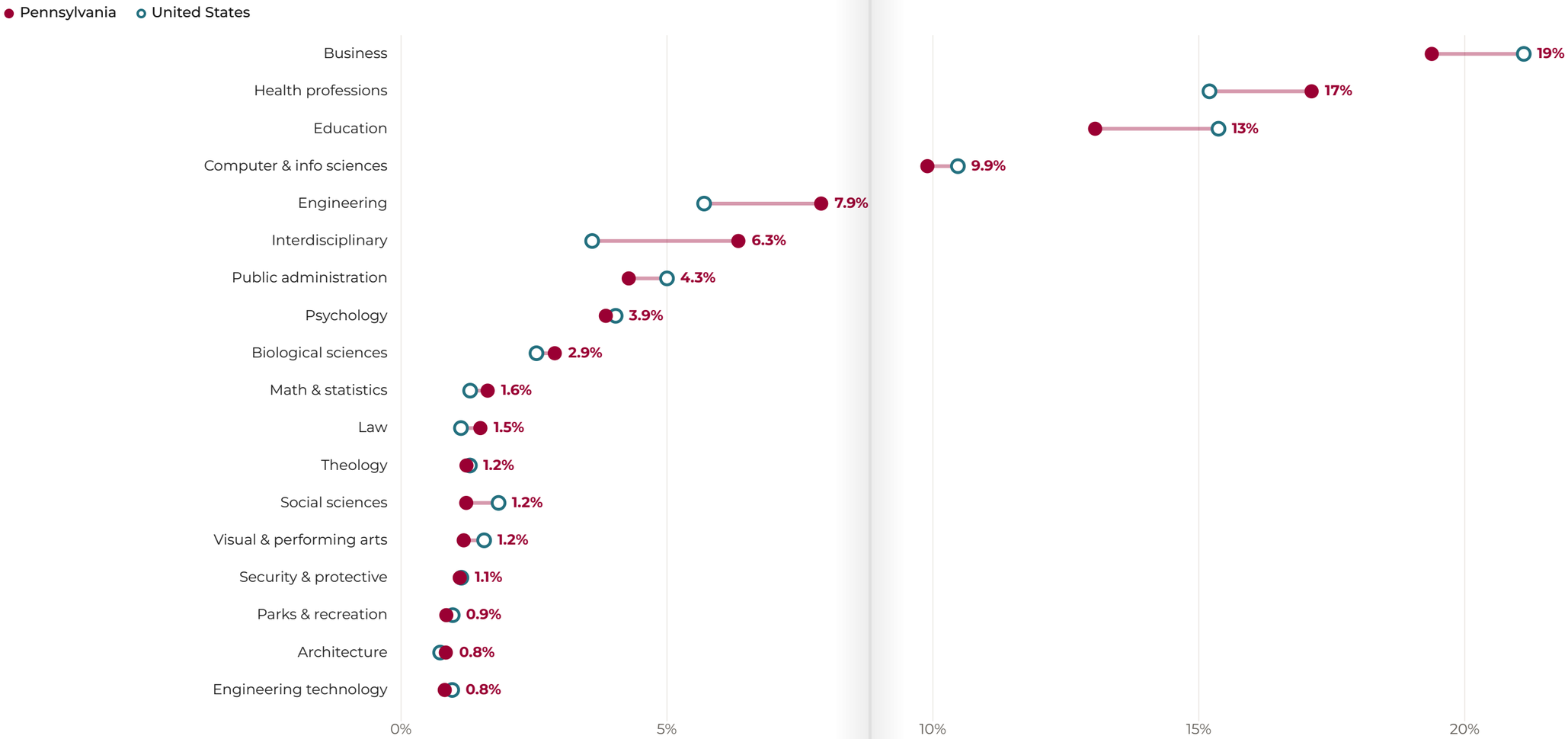
Relative to the nation, Pennsylvania's bachelor's degrees are most concentrated in physical sciences (1.50× the U.S. share), math & statistics (1.30× the U.S. share), engineering (1.24× the U.S. share).

**6.9%**
**Bachelor's in computer science, 2025**
U.S. **6.9%** · **18th** of 51 · +3.5 pp since 2015

**8.1%**
**Bachelor's in engineering, 2025**
U.S. **6.8%** · **13th** of 51 · +1.1 pp since 2015

**17%**
**Bachelor's in business, 2025**
U.S. **19%** · **39th** of 51 · −1.2 pp since 2015

## Figure 29 What Do Pennsylvania's Master's Graduates Study?




Share of master's degrees by field, 2025; largest fields. Data from NCES, IPEDS.

Pennsylvania's largest master's fields are business (19%; U.S. 21%), health professions (17%; U.S. 15%), education (13%; U.S. 15%).

International students earn 24% of all master's degrees in the state (U.S. 22%), and STEM fields account for 25% of them.

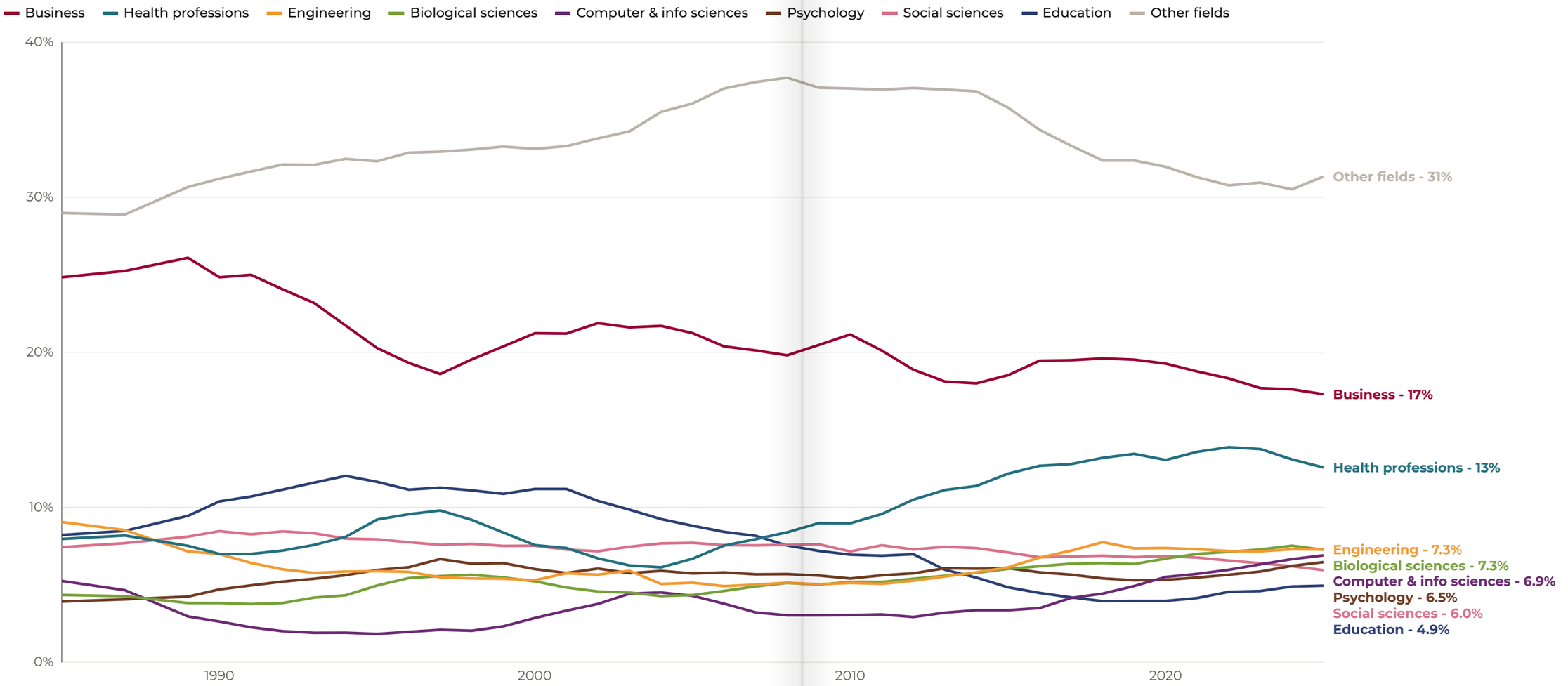
**25%**
**STEM share of master's, 2025**
U.S. **23%** · **15th** of 51 · +7.0 pp since 2015

**24%**
**Int. share of all master's, 2025**
U.S. **22%** · **12th** of 51 · +7.2 pp since 2015

**62%**
**Int. share of CS master's, 2025**
U.S. **65%** · **27th** of 47 · −1.1 pp since 2015

PENNSYLVANIA · STATE TALENT PROFILE

## Figure 30 How Has Pennsylvania's Bachelor's Field Mix Changed?




Share of bachelor's degrees in the largest fields. Data from NCES, IPEDS Completions.

Since 2015, computer & info sciences has gained the most ground, 3.5 points of the state's bachelor's degrees, while business lost the most (1.2 points).

The chart follows the share of bachelor's degrees in each of the state's largest fields; smaller fields are grouped as other fields.

FIELDS OF STUDY

# Where Is Pennsylvania Specialized, and What Do Its Doctoral Graduates Study?

The location quotient divides a field's share of Pennsylvania's bachelor's degrees by its share of all U.S. bachelor's degrees in 2025. A quotient of 1.00 means the same share as the nation; 2.00 means twice the national share. The table lists the ten fields with the highest quotients among fields with at least 50 degrees.

**Highest location quotients.** Physical sciences 1.50 (1.9% of the state's bachelor's degrees; U.S. 1.3%); math & statistics 1.30 (1.5% of the state's bachelor's degrees; U.S. 1.2%); engineering 1.24 (7.3% of the state's bachelor's degrees; U.S. 5.9%).

## Table 5 Where Is Pennsylvania Most Specialized?

| Field | Location quotient | State share | U.S. share | Degrees |
|---|---|---|---|---|
| Physical sciences | **1.50** | 1.9% | 1.3% | 1.56K |
| Math & statistics | **1.30** | 1.5% | 1.2% | 1.24K |
| Engineering | **1.24** | 7.3% | 5.9% | 5.9K |
| Architecture | **1.23** | 0.7% | 0.6% | 564 |
| Philosophy & religion | **1.21** | 0.7% | 0.6% | 549 |
| Interdisciplinary | **1.17** | 3.4% | 2.9% | 2.72K |
| English | **1.15** | 1.7% | 1.5% | 1.4K |
| History | **1.14** | 1.2% | 1.0% | 956 |
| Education | **1.14** | 4.9% | 4.3% | 4.01K |
| Biological sciences | **1.10** | 7.3% | 6.6% | 5.89K |

Bachelor's degrees; location quotient = state share ÷ U.S. share; fields with 50+ degrees. Data from NCES, IPEDS.





**27%**
STEM share of bachelor's, 2025
U.S. **24%** · **10th** of 51

**6.9%**
Bachelor's in computer science, 2025
U.S. **6.9%** · **18th** of 51

**8.1%**
Bachelor's in engineering, 2025
U.S. **6.8%** · **13th** of 51

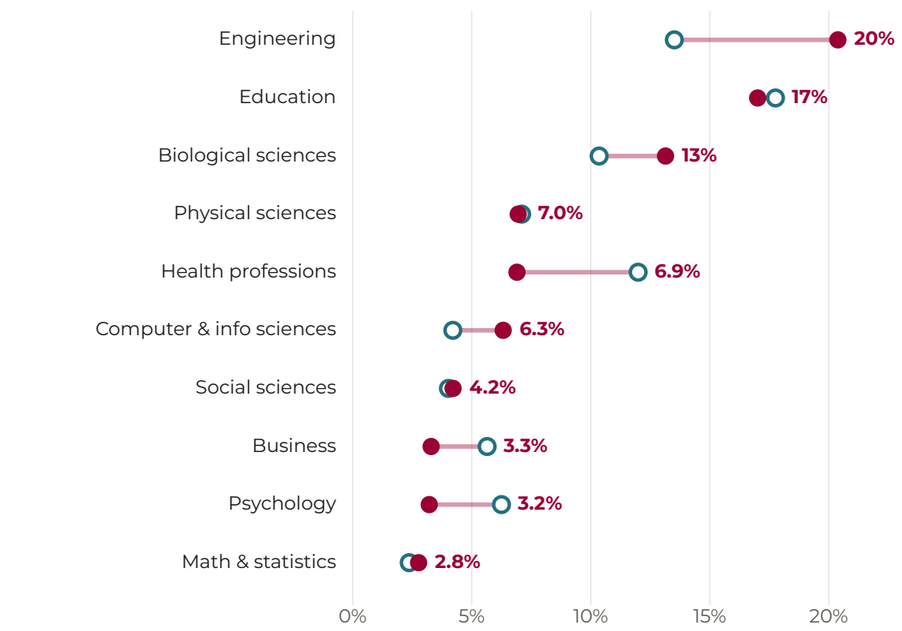
**17%**
Bachelor's in business, 2025
U.S. **19%** · **39th** of 51

Each row is a field's share of the research doctorates awarded in Pennsylvania in 2025 (filled dot), against its share nationally (open dot), for the ten largest fields with at least 20 doctorates. Research doctorates exclude professional doctorates such as the M.D. and J.D.

**Largest doctoral fields.** Engineering 20% (U.S. 14%); education 17% (U.S. 18%); biological sciences 13% (U.S. 10%).

**STEM share of research doctorates.** 50% in 2025 (U.S. 39%; 10th of 51 states), up 2.3 points from 48% in 2024, and up 0.3 points since 2015 (50%).

## Figure 31 What Do Pennsylvania's Doctoral Graduates Study?



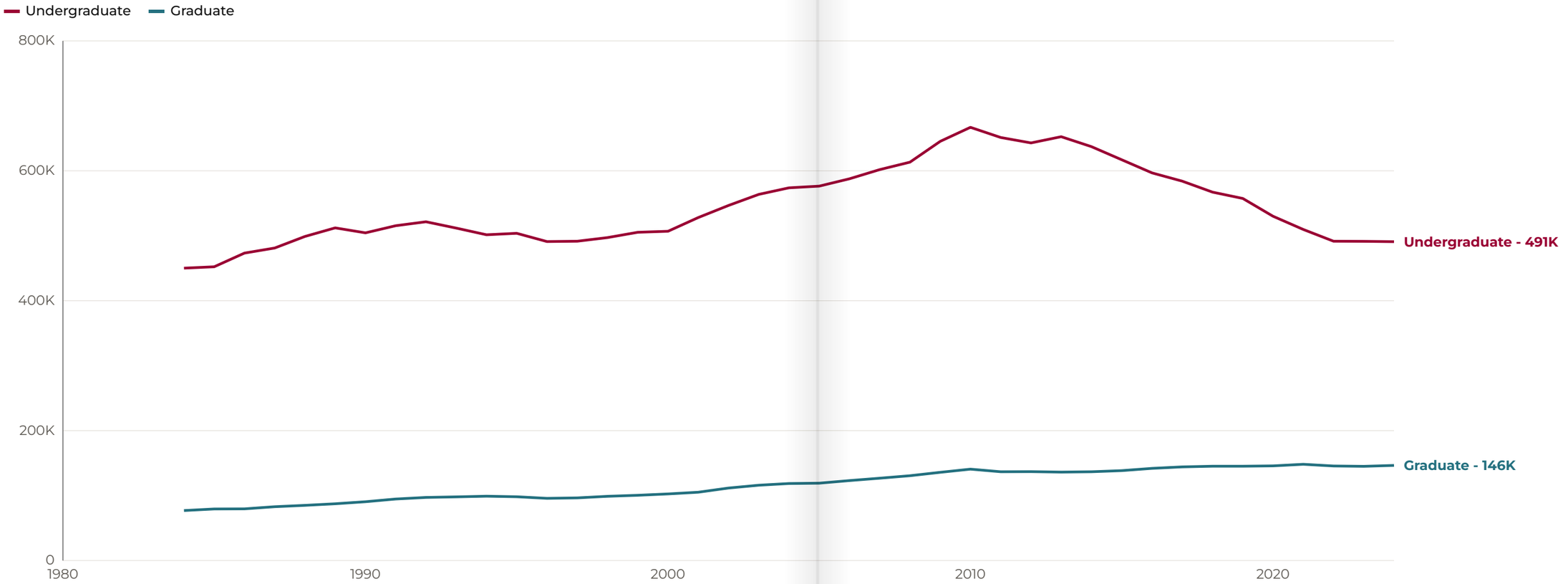

Share of research doctorates by field, 2025; fields with 20+ doctorates. Data from NCES, IPEDS.

PENNSYLVANIA · STATE TALENT PROFILE

SECTION 13

# Students and Money

## Figure 32 How Has Enrollment Changed?




Fall headcount enrollment. Data from NCES, IPEDS Fall Enrollment.

In fall 2024, Pennsylvania enrolled 491K undergraduates and 146K graduate students, 637K in all (7th of 51).

Undergraduate enrollment is down 20% since 2015 and graduate enrollment up 6% since 2015. Graduate students are 23% of all students (U.S. 17%).

**637K**

**Students enrolled, 2024**

U.S. **19.7M** · **7th** of 51 · −16% since 2015

**23%**

**Graduate share of students, 2024**

U.S. **17%** · **4th** of 51 · +4.6 pp since 2015

**27%**

**Students attending part-time, 2024**

U.S. **39%** · **48th** of 51 · −2.1 pp since 2015

STUDENTS AND MONEY

# Who Enrolls in Pennsylvania, and Where?

In fall 2024, White students were the largest group at Pennsylvania's degree-granting institutions, **56%** of enrollment against 45% nationally, down 7.2 points since 2015.

Black students were **11%** (U.S. 12%), unchanged since 2015; Hispanic students were **9.6%** (U.S. 21%), up 3.6 points since 2015.

Asian and Pacific Islander students were **7.5%** (U.S. 7.4%), up 2.5 points since 2015, and International students were **7.1%** (U.S. 5.3%), up 1.0 point since 2015. "International" counts U.S. nonresidents on temporary visas, whatever their race.

## Figure 33 Who Enrolls in Pennsylvania?

■ Pennsylvania ■ United States

| | | Change since 2015 |
|---|---|---|
| White | 56% · U.S. 45% | −7.2 pts |
| Black | 11% · U.S. 12% | −0.0 pts |
| Hispanic | 9.6% · U.S. 21% | +3.6 pts |
| Asian / Pacific Islander | 7.5% · U.S. 7.4% | +2.5 pts |
| International | 7.1% · U.S. 5.3% | +1.0 pts |
| Two or more | 3.6% · U.S. 4.2% | +1.1 pts |
| American Indian | 0.2% · U.S. 0.6% | −0.0 pts |

Share of fall 2024 headcount enrollment by race and ethnicity; students of unknown race are left out of the chart. Data from NCES, IPEDS.





**56%**
White students
U.S. **45%** · −7.2 pp since 2015

**11%**
Black students
U.S. **12%** · −0.0 pp since 2015

**45%**
At private nonprofit institutions
+3.3 pp since 2015

**39%**
At public 4-year institutions
+1.6 pp since 2015

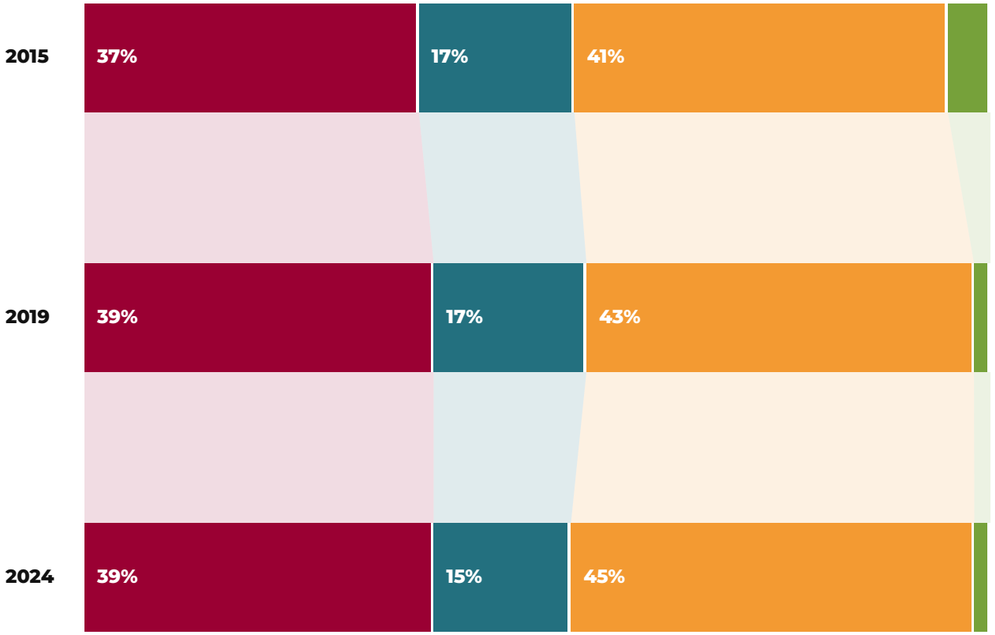
Private nonprofit institutions enrolled **45%** of students, up 3.3 points from 41% in 2015 in fall 2024. Public 4-year institutions enrolled **39%** of students, up 1.6 points from 37% in 2015.

Public 2-year institutions enrolled **15%** of students, down 2.0 points from 17% in 2015, and for-profit institutions enrolled **1.8%** of students, down 2.9 points from 4.7% in 2015.

Public institutions together enrolled **54%** of students, against 73% nationally, 43rd among the states.

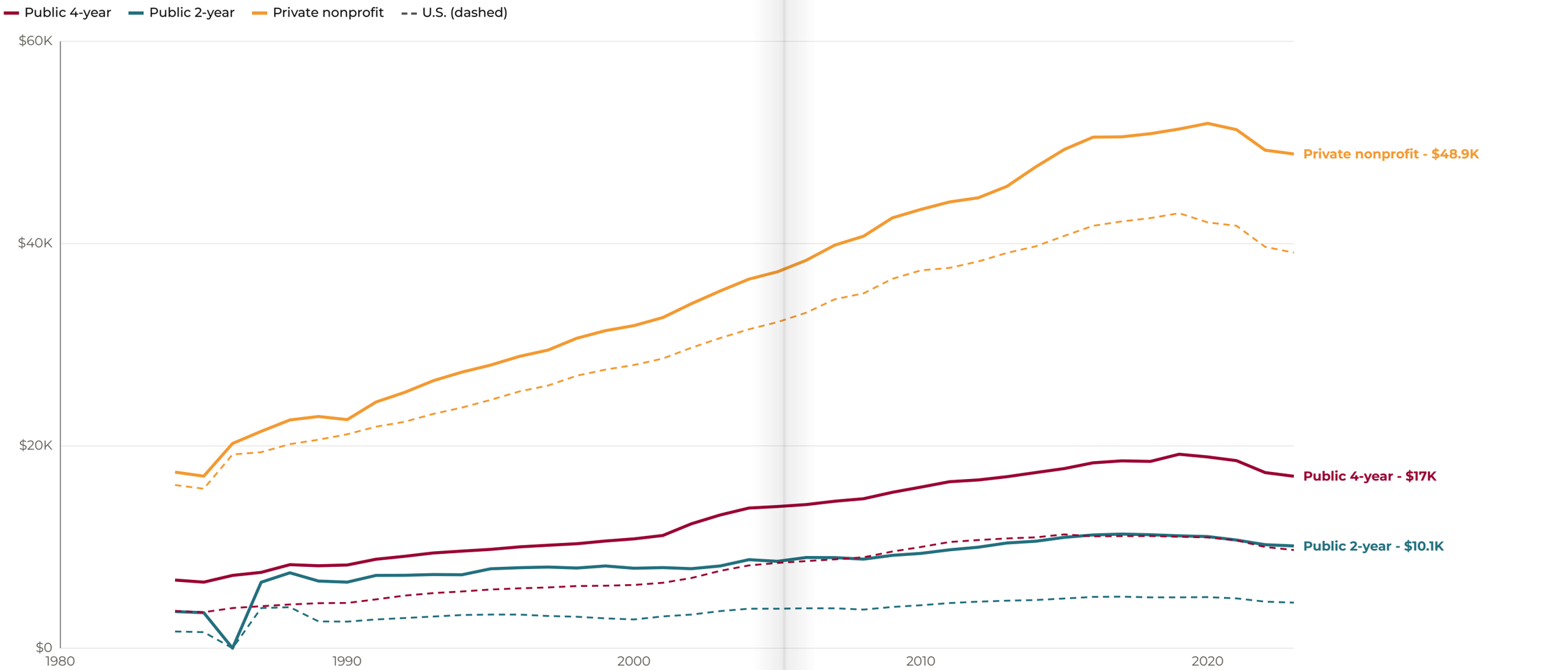
## Figure 34 Where Do Students Enroll?

■ Public 4-year ■ Public 2-year ■ Private nonprofit ■ For-profit




Share of fall headcount enrollment by sector, 2015, 2019, 2024. Data from NCES, IPEDS.

## Figure 35 What Does In-State Tuition Cost?




Published in-state tuition and fees, 2024 dollars (CPI-U); state solid, U.S. dashed. Data from NCES, IPEDS.

In-state tuition and fees at Pennsylvania's public four-year institutions are $17K (2023, 2024 dollars), 75% above the U.S. ($9.71K); the state ranks 3rd of 51.

State and local appropriations come to $7.14K per full-time-equivalent student at public institutions (46th of 51), and tuition makes up 72% of their core revenue (U.S. 43%).

**$17K**
**In-state tuition, public 4-year, 2023**
U.S. **$9.71K** · **3rd** of 51 · −4.3% since 2015

**$7.14K**
**Appropriations per student, 2024**
U.S. **$11.4K** · **46th** of 51 · +20% since 2015

**72%**
**Tuition share of public revenue, 2024**
U.S. **43%** · **3rd** of 51 · −4.0 pp since 2015

STUDENTS AND MONEY

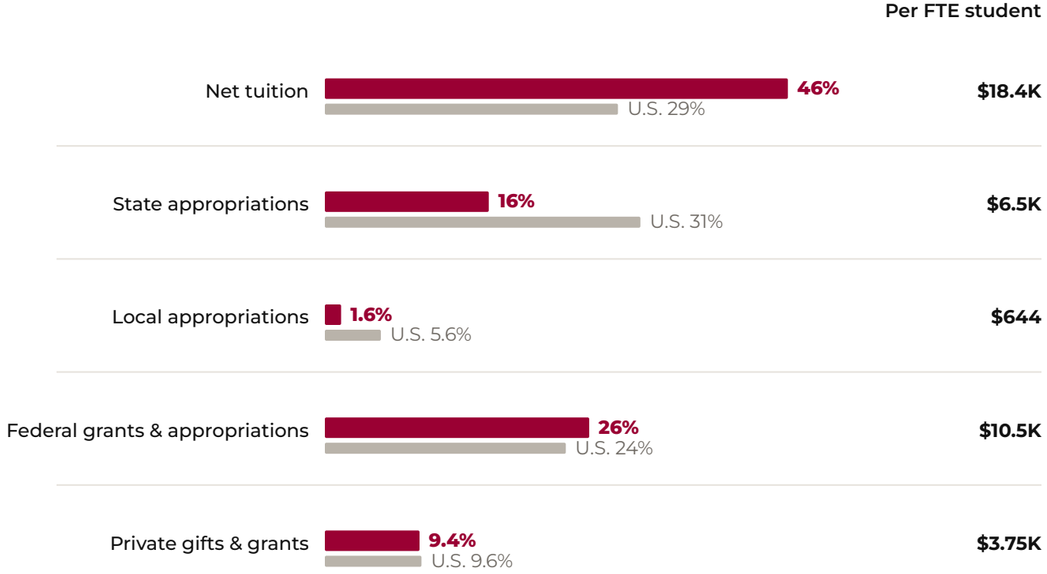
# How Are Pennsylvania's Public Institutions Funded, and What Do They Produce?

The bars give each source's share of the core revenue of Pennsylvania's public institutions, with the nation for comparison, and at right the dollars per full-time-equivalent (FTE) student it provides. Values are in 2024 dollars (CPI-U). Net tuition is tuition and fees after institutional discounts and allowances.

**Net tuition per FTE student.** $18.4K in FY2024 (U.S. $8.96K), down 0.6% from $18.5K in FY2023, and down 2.9% since FY2015 ($18.9K).

**State appropriations per FTE student.** $6.5K in FY2024 (U.S. $9.64K), up 5.0% from $6.19K in FY2023, and up 19% since FY2015 ($5.46K).

## Figure 36 How Are Public Institutions Funded?




Share of core revenue per FTE student by source, public institutions, FY2024; dollars per FTE at right, 2024 dollars (CPI-U). Data from NCES, IPEDS Finance.





**$17K**
In-state tuition, public 4-year, 2023
U.S. **$9.71K** · **3rd** of 51

**$7.14K**
Appropriations per student, 2024
U.S. **$11.4K** · **46th** of 51

**72%**
Tuition share of public revenue, 2024
U.S. **43%** · **3rd** of 51

**68%**
Six-year graduation rate, 2024
U.S. **65%** · **15th** of 51

The table brings together five measures of what public higher education costs and what it produces in Pennsylvania: in-state tuition at public four-year institutions, state and local appropriations per full-time-equivalent student, tuition's share of public institutions' core revenue, the six-year bachelor's graduation rate, and the average full-time faculty salary.

**In-state tuition, public 4-year.** $17K in 2023 (U.S. $9.71K; 3rd of 51 states), down 2.1% from $17.4K in 2022, and down 4.3% since 2015 ($17.8K).

## Table 6 Money and Results

| Measure | Year | Pennsylvania | U.S. | Rank | Change since 2015 |
|---|---|---|---|---|---|
| In-state tuition, public 4-year | 2023 | **$17K** | $9.71K | 3rd of 51 | −4.3% |
| Appropriations per student | 2024 | **$7.14K** | $11.4K | 46th of 51 | +20% |
| Tuition share of public revenue | 2024 | **72%** | 43% | 3rd of 51 | −4.0 pp |
| Six-year graduation rate | 2024 | **68%** | 65% | 15th of 51 | +0.9 pp |
| Average faculty salary | 2024 | **$109K** | $106K | 14th of 51 | −8.2% |

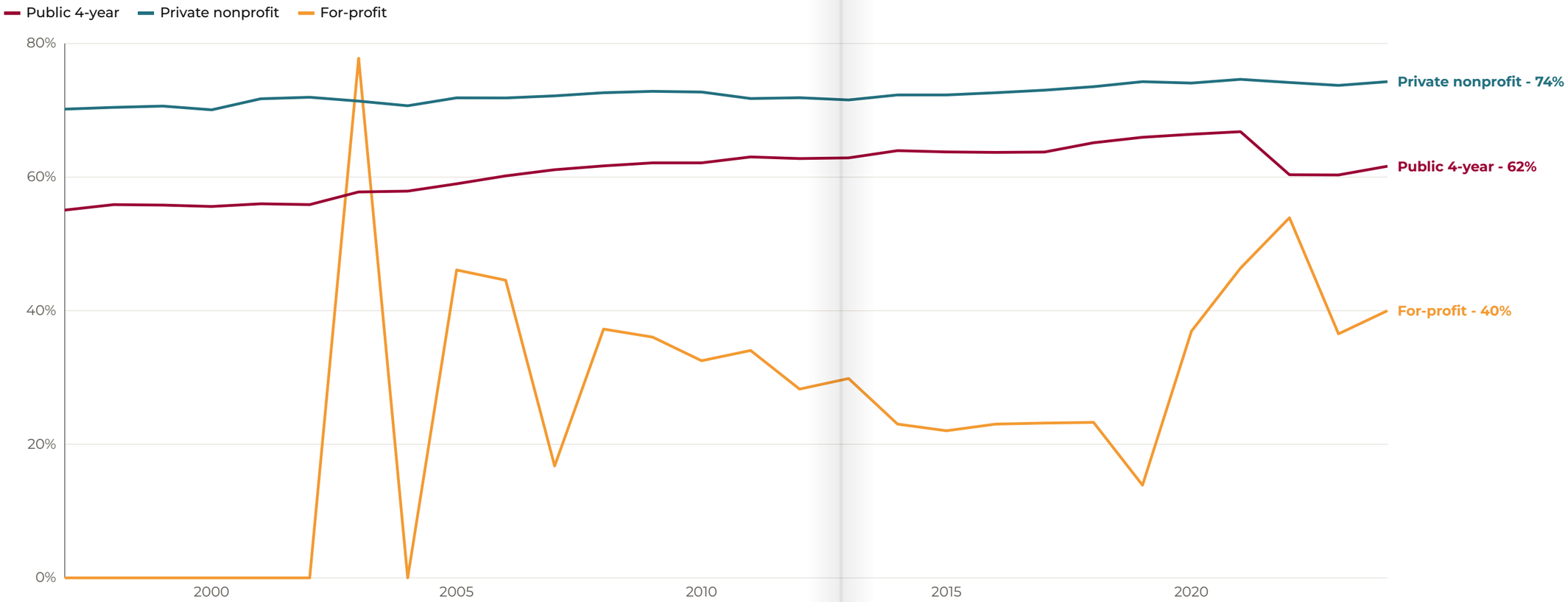
Rank among the 50 states and DC (1 = highest). Dollars in 2024 dollars (CPI-U). Data from NCES, IPEDS.

PENNSYLVANIA · STATE TALENT PROFILE

SECTION 14

# Student Outcomes

## Figure 37 How Do Graduation Rates Differ by Sector?




Six-year bachelor's graduation rate by sector of institution. Data from NCES, IPEDS Graduation Rates.

68% of students who started a bachelor's degree at a four-year institution in Pennsylvania graduated within six years (2024), 3.0 points above the U.S. (65%); the state ranks 15th of 51.

By sector: public 4-year 62%, private nonprofit 74%, for-profit 40%. At two-year institutions, 42% complete within 150% of normal time (U.S. 41%).

**68%**
**Six-year graduation rate, 2024**
U.S. **65%** · **15th** of 51 · +0.9 pp since 2015

**42%**
**Two-year completion rate, 2024**
U.S. **41%** · **22nd** of 50 · +5.4 pp since 2015

**16.5**
**Bachelor's per 100 undergraduates, 2025**
U.S. **12.3** · **6th** of 51 · +1.8 since 2015

STUDENT OUTCOMES

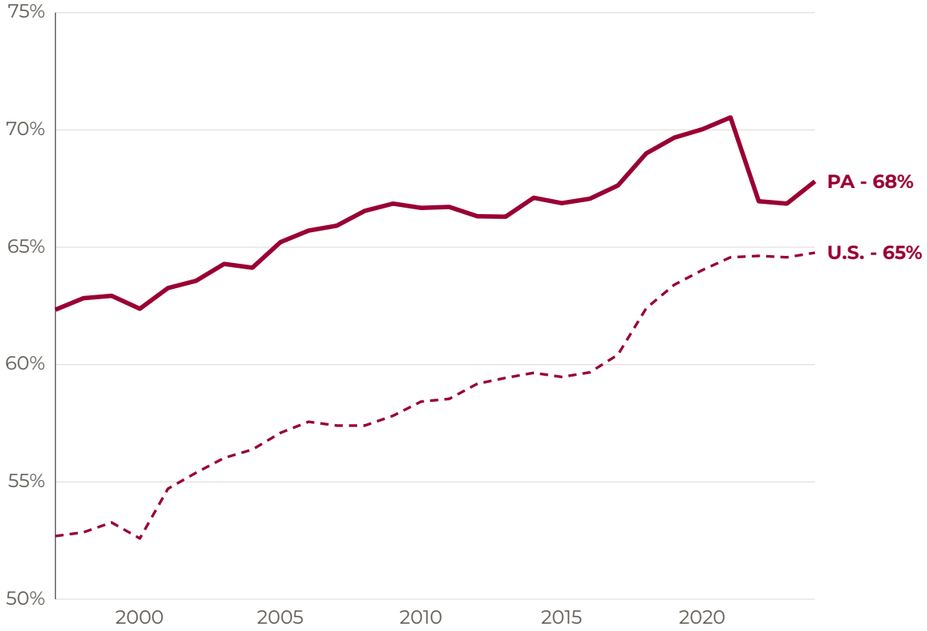
# How Many of Pennsylvania's Students Graduate?

The line follows the share of first-time, full-time bachelor's-degree students at Pennsylvania's four-year institutions who graduated from the institution where they started within six years (150% of normal time). Each year is the cohort that entered six years earlier. The state is solid and the United States dashed.

**Six-year graduation rate.** 68% in 2024 (U.S. 65%), up 1.0 point from 67% in 2023, and up 0.9 points since 2015 (67%).

**Public 4-year institutions.** 62% in 2024, up 1.3 points from 60% in 2023, and down 2.2 points since 2015 (64%).

## Figure 38 Six-Year Bachelor's Graduation Rate




Four-year institutions; state solid, U.S. dashed. Data from NCES, IPEDS Graduation Rates.





**68%**
Six-year graduation rate, 2024
U.S. **65%** · **15th** of 51

**42%**
Two-year completion rate, 2024
U.S. **41%** · **22nd** of 50

**16.5**
Bachelor's per 100 undergraduates, 2025
U.S. **12.3** · **6th** of 51

Each row is the six-year bachelor's graduation rate for one racial or ethnic group of students at Pennsylvania's four-year institutions (filled dot), against the national rate for the same group (open dot). Groups with fewer than 100 students in the cohort are left out.

**White.** 71% in 2024 (U.S. 68%).

**Hispanic.** 62% in 2024 (U.S. 59%).

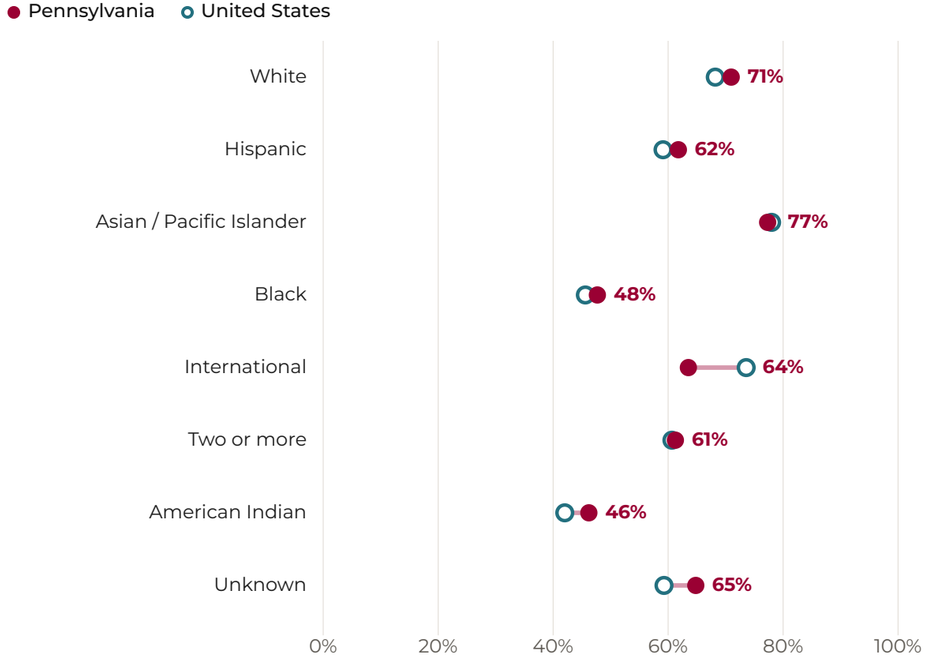
**Asian / Pacific Islander.** 77% in 2024 (U.S. 78%).

**Black.** 48% in 2024 (U.S. 46%).

**International.** 64% in 2024 (U.S. 74%).

**Two or more.** 61% in 2024 (U.S. 61%).

## Figure 39 How Do Graduation Rates Differ by Race and Ethnicity?



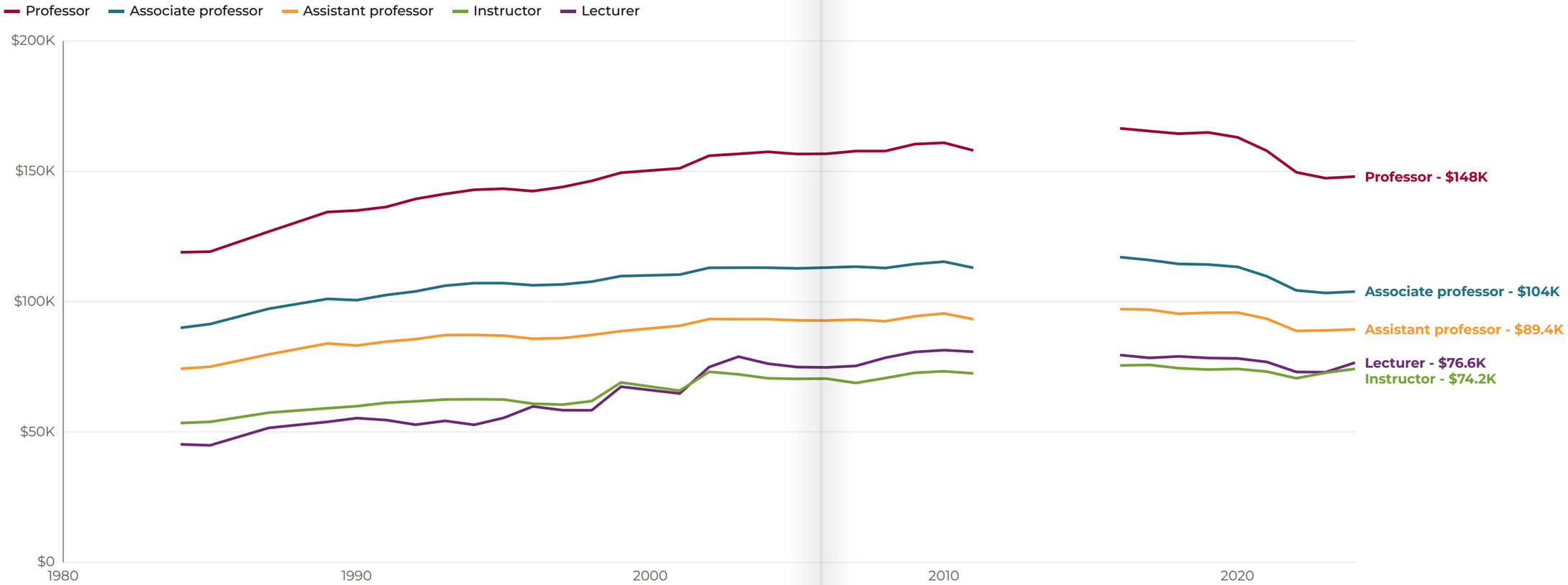

Six-year rate, 2024; groups with 100+ students. Data from NCES, IPEDS.

PENNSYLVANIA · STATE TALENT PROFILE

SECTION 15

# Faculty

## Figure 40 What Do Pennsylvania's Faculty Earn, by Rank?




Average salary of full-time instructional faculty on 9/10-month contracts, by academic rank, 2024 dollars (CPI-U). Data from NCES, IPEDS Human Resources.

Full-time instructional faculty on 9/10-month contracts at Pennsylvania's institutions earned an average of $109K in 2024 (2024 dollars), 3% above the U.S. ($106K); the state ranks 14th of 51. By rank (2024): professors $148K, associate professors $104K, assistant professors $89.4K, instructors $74.2K, lecturers $76.6K.

Pennsylvania's institutions had 34.6K full-time instructional staff in 2024, down 6% since 2015. Average salary across all ranks is down 8% since 2016.

**$109K**

**Average faculty salary, 2024**

U.S. **$106K** · **14th** of 51 · −8.2% since 2016

FACULTY

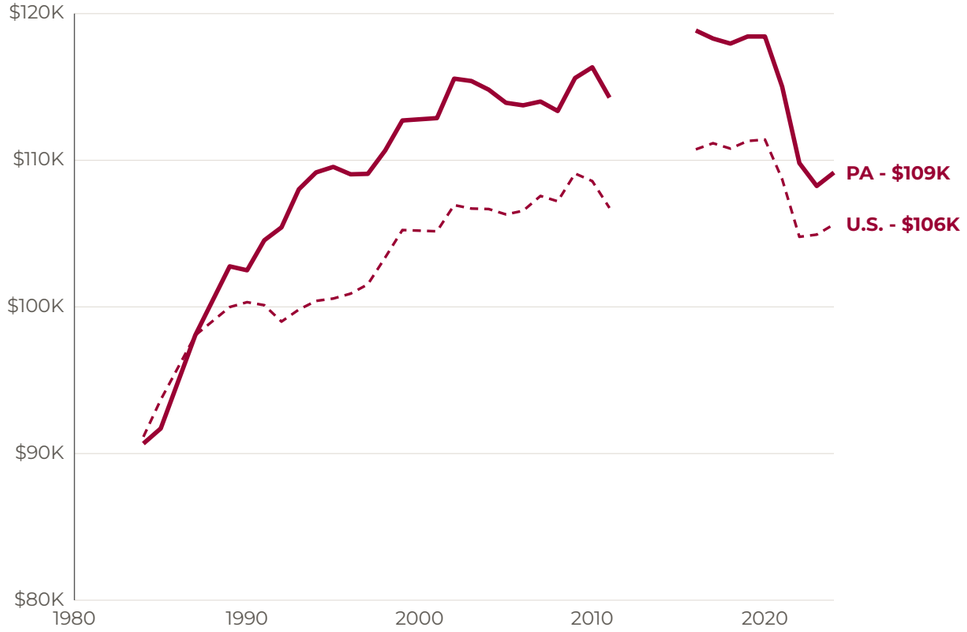
# What Do Pennsylvania's Faculty Earn, and How Many Hold Tenure?

The line follows the average salary of full-time instructional faculty on 9- or 10-month contracts at Pennsylvania's degree-granting institutions, all ranks combined, in 2024 dollars (CPI-U). The state is solid and the United States dashed.

**Average salary.** $109K in 2024 (U.S. $106K), up 0.9% from $108K in 2023, and down 8.1% since 2016 ($119K).

**State rank on average salary.** $109K in 2024 (U.S. $106K; 14th of 51 states), up 0.9% from $108K in 2023, and down 8.2% since 2016 ($119K).

## Figure 41 Average Faculty Salary




Full-time instructional faculty, 9/10-month, 2024 dollars. Data from NCES, IPEDS.





**$109K**
Average faculty salary, 2024
U.S. **$106K** · **14th** of 51

**34.6K**
Full-time instructional staff, 2024

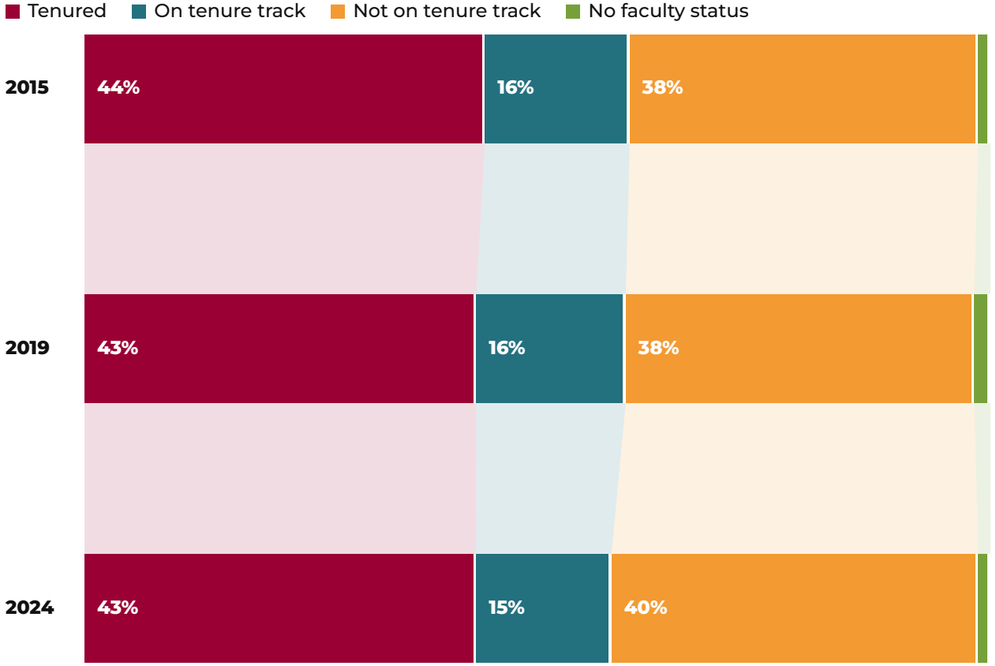
**43%**
Tenured, 2024

The bars divide Pennsylvania's full-time instructional staff by tenure status in 2015, 2019 and 2024: tenured, on the tenure track, not on the tenure track (for example, lecturers on fixed-term contracts), and staff at institutions with no tenure system.

**Tenured, share of full-time instructional staff.** 43% in 2024, up 0.3 points from 43% in 2023, and down 1.0 point since 2015 (44%).

**On tenure track, share of full-time instructional staff.** 15% in 2024, down 0.6 points from 16% in 2023, and down 1.1 points since 2015 (16%).

## Figure 42 Tenure Status

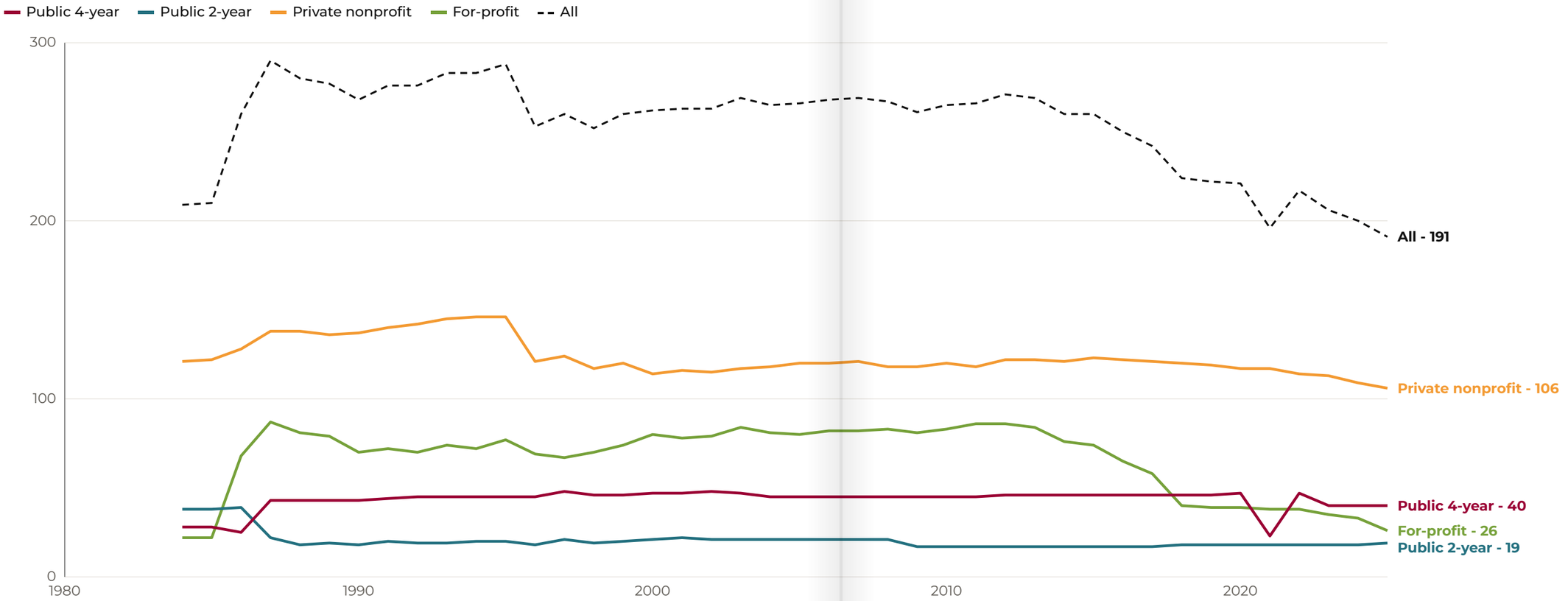



Full-time instructional staff by tenure status, 2015, 2019, 2024. Data from NCES, IPEDS.

SECTION 16

# Institutions

## Figure 43 How Many Institutions Does Pennsylvania Have?




Degree-granting institutions by sector. Data from NCES, IPEDS.

Pennsylvania has 189 degree-granting institutions: 40 public 4-year, 19 public 2-year, 106 private nonprofit, 26 for-profit (2025).

The number of institutions is down 27% since 2015. 54% of students attend public institutions and 1.8% attend for-profit ones.

**54%**

**Students at public institutions, 2024**

U.S. **73%** · **43rd** of 51 · −0.4 pp since 2015

**1.8%**

**Students at for-profit institutions, 2024**

U.S. **5.5%** · **26th** of 51 · −2.9 pp since 2015

**17%**

**Students fully online, 2024**

U.S. **27%** · **46th** of 51 · +5.5 pp since 2015

INSTITUTIONS

# Which Institutions Lead in Pennsylvania?

Institutions in Pennsylvania ranked by the STEM master's and research doctorates they awarded to international students in 2025. The share columns give international students' share of each institution's STEM degrees at that level.

**Largest.** Carnegie Mellon University (2.21K international STEM master's, 193 doctorates); University of Pennsylvania (1.21K international STEM master's, 122 doctorates); Pennsylvania State University-Main Campus (319 international STEM master's, 269 doctorates).

## Table 7 Which Institutions Train the Most International STEM Graduates?

| Institution | Intl STEM master's | Int. share | Intl STEM doctorates | Int. share |
|---|---|---|---|---|
| Carnegie Mellon University<br>Pittsburgh · Private nonprofit | **2.21K** | 74% | 193 | 55% |
| University of Pennsylvania<br>Philadelphia · Private nonprofit | **1.21K** | 52% | 122 | 35% |
| Pennsylvania State University-Main Campus<br>University Park · Public 4-year | **319** | 50% | 269 | 51% |
| Harrisburg University of Science and Technology<br>Harrisburg · Private nonprofit | **454** | 97% | 4 | 40% |
| University of Pittsburgh-Pittsburgh Campus<br>Pittsburgh · Public 4-year | **236** | 39% | 132 | 46% |
| Drexel University<br>Philadelphia · Private nonprofit | **258** | 37% | 39 | 27% |
| Gannon University<br>Erie · Private nonprofit | **271** | 95% | – | – |
| Robert Morris University<br>Moon Township · Private nonprofit | **208** | 89% | 4 | 31% |

Degrees awarded in 2025. Data from NCES, IPEDS.





**189**
Degree-granting institutions

**54%**
Students at public institutions, 2024
U.S. **73%** · **43rd** of 51

**1.8%**
Students at for-profit institutions, 2024
U.S. **5.5%** · **26th** of 51

**17%**
Students fully online, 2024
U.S. **27%** · **46th** of 51

Pennsylvania's largest degree-granting institutions by fall 2024 headcount enrollment, with the international share of graduate students, STEM bachelor's degrees awarded in 2025, the six-year graduation rate and the admission rate (a dash where the institution admits all applicants or does not report one).

**Largest.** Pennsylvania State University-Main Campus (50.7K students); University of Pittsburgh-Pittsburgh Campus (35.7K students); Temple University (29.6K students).

**Degree-granting institutions.** 189.

## Table 8 What Are Pennsylvania's Largest Institutions?

| Institution | Enrollment | Int. share, grad | STEM bachelor's | Six-year grad rate | Admit rate |
|---|---|---|---|---|---|
| Pennsylvania State University-Main Campus | **50.7K** | 44% | 4.16K | 86% | 61% |
| University of Pittsburgh-Pittsburgh Campus | **35.7K** | 22% | 2.2K | 85% | 58% |
| Temple University | **29.6K** | 13% | 1.14K | 75% | 80% |
| University of Pennsylvania | **29.1K** | 30% | 950 | 97% | 5.4% |
| Drexel University | **21.2K** | 15% | 1.04K | 78% | 79% |
| West Chester University of Pennsylvania | **17.2K** | 1.9% | 263 | 71% | 78% |
| Carnegie Mellon University | **15.9K** | 62% | 1.23K | 94% | 12% |
| Pennsylvania State University-World Campus | **13.8K** | 3.5% | 302 | 34% | 91% |
| Community College of Philadelphia | **12.9K** | – | – | – | – |
| Harrisburg Area Community College | **12.6K** | – | – | – | – |

Fall 2024 enrollment; degrees 2025. Data from NCES, IPEDS.

NOTES

# About These Data

**International students**

U.S. nonresidents on temporary visas, as IPEDS defines them. IPEDS does not record country of citizenship. Permanent residents count with U.S. citizens.

**STEM**

CIP families 03, 04, 11, 14, 15, 26, 27 and 40. Doctorates are research and scholarship doctorates; professional doctorates appear only in the degree-count figure. Degrees are first majors only.

**Universe**

Degree-granting institutions located in the state; Title IV participants from 1996. Geography is the institution's location, not the student's home. Enrollment is fall headcount.

**Migration**

First-time degree-seeking undergraduates by home state, reported in even years before 2001 and every year since; odd years with incomplete reporting are omitted. Net migration is students arriving from other states minus residents leaving, per 100 resident first-time students.

**Money**

All dollar values are in 2024 dollars using the CPI-U. Appropriations and revenue are per full-time-equivalent student at public institutions.

**Race and ethnicity**

IPEDS did not collect race and ethnicity in every year before 2010, and categories changed in 2008–2010; “Two or more” exists only from then.

**Countries of citizenship**

From DHS SEVIS by the Numbers: active F-1 and M-1 student records in the month named, at every level from primary school to doctorate, by the state of the school. IPEDS records only whether a student is a U.S. nonresident, not where they come from. SEVIS counts are monthly snapshots and are not directly comparable with IPEDS fall enrollment or degree counts.

**Peers**

Comparison states are the four closest in total fall enrollment.

**Source**

National Center for Education Statistics, Integrated Postsecondary Education Data System (IPEDS): Institutional Characteristics, Fall Enrollment, Completions, Graduation Rates, Finance, Human Resources and Admissions components, 1980–2025, harmonized into a state panel. Department of Homeland Security, SEVIS by the Numbers data mapping tool, 2014–2026. Generated November 2026.



PENNSYLVANIA · STATE TALENT PROFILE

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# Pennsylvania – By the Numbers

| Measure | 2015 | Latest | Change since 2015 | U.S., latest | U.S. change since 2015 | Rank | Source |
|---|---|---|---|---|---|---|---|
| **K-12 PREPARATION** | | | | | | | |
| Grade 8 mathematics, average score | 284 | **276**[2024] | −7.4 ▼ | 272 | −9.0 ▼ | #18 | [5] |
| Grade 8 mathematics: at or above Proficient | 36% | **31%**[2024] | −4.6 pp ▼ | 27% | −5.3 pp ▼ | #15 | [5] |
| Grade 4 mathematics, average score | 243 | **238**[2024] | −5.1 ▼ | 237 | −2.5 ▼ | #23 | [5] |
| Grade 4 mathematics: at or above Proficient | 45% | **41%**[2024] | −3.6 pp ▼ | 39% | 0.0 pp | #18 | [5] |
| Grade 8 reading, average score | 269 | **259**[2024] | −9.6 ▼ | 257 | −7.3 ▼ | #19 | [5] |
| Grade 8 reading: at or above Proficient | 39% | **31%**[2024] | −8.3 pp ▼ | 29% | −4.0 pp ▼ | #18 | [5] |
| Grade 4 reading, average score | 227 | **216**[2024] | −10.9 ▼ | 214 | −7.1 ▼ | #17 | [5] |
| Grade 8 science, average score, 2009 framework | – | **151**[2011] | – | 151 | – | #30 | [5] |
| Current spending per public-school pupil | $18.8K | **$21K**[2021] | +11% ▲ | $16.8K | +12% ▲ | #13 | [3] |
| Public high school 4-year graduation rate, ACGR | 85% | **87%**[2022] | +2.0 pp ▲ | 87% | +4.0 pp ▲ | #19 | [3] |
| Public high school graduates | 125K | **127K**[2024] | +1.8% ▲ | 3.49M | +8.8% ▲ | #6 | [3] |
| Average public-school teacher salary | $84.3K | **$79.3K**[2022] | −5.9% ▼ | $72.9K | −3.0% ▼ | #12 | [3] |
| Pupils per teacher, public schools | 14.2 | **13.3**[2022] | −0.9 ▼ | 15.4 | −0.6 ▼ | #36 | [3] |
| Public elementary and secondary enrollment | 1.72M | **1.69M**[2024] | −1.5% ▼ | 49.4M | −2.1% ▼ | #7 | [3] |
| Public-school students eligible for free or reduced-price lunch | 46% | **57%**[2023] | +11.0 pp ▲ | 53% | +1.5 pp ▲ | #15 | [3] |
| English learners as a share of public-school students | 3.0% | **4.6%**[2022] | +1.6 pp ▲ | 11% | +1.0 pp ▲ | #40 | [3] |
| Charter-school share of public-school enrollment | 7.7% | **9.7%**[2023] | +2.0 pp ▲ | 7.6% | +2.2 pp ▲ | #12 | [3] |
| High schools offering calculus | 79%[2016] | **74%**[2024] | −5.5 pp ▼ | 51% | −5.8 pp ▼ | #2 | [14] |
| High schools offering physics | 83%[2016] | **84%**[2024] | +0.2 pp ▲ | 64% | −2.4 pp ▼ | #6 | [14] |
| High schools offering chemistry | 91%[2016] | **90%**[2024] | −0.9 pp ▼ | 81% | +0.7 pp ▲ | #12 | [14] |
| High schools offering Algebra II | 93%[2016] | **91%**[2024] | −2.2 pp ▼ | 87% | +1.3 pp ▲ | #21 | [14] |
| High schools offering computer science | 66%[2021] | **65%**[2024] | −1.5 pp ▼ | 55% | +2.0 pp ▲ | #23 | [14] |
| High schools offering any AP course | 79%[2016] | **79%**[2024] | −0.2 pp ▼ | 56% | 0.0 pp | #7 | [14] |
| High schools offering AP computer science | 28%[2021] | **37%**[2024] | +8.5 pp ▲ | 25% | +2.6 pp ▲ | #12 | [14] |

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

## PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

| Measure | 2015 | Latest | Change since 2015 | U.S., latest | U.S. change since 2015 | Rank | Source |
|---|---|---|---|---|---|---|---|
| **K-12 PREPARATION (CONTINUED)** | | | | | | | |
| High schools offering dual enrollment | 60%[2016] | **74%[2024]** | +14.1 pp ▲ | 71% | +11.1 pp ▲ | #31 | [14] |
| AP course enrollments per 100 high-school students | 14.1[2016] | **14.2[2024]** | 0.0 | 18.4 | +1.0 ▲ | #29 | [14] |
| Calculus enrollments per 100 high-school students | 5.6[2016] | **3.9[2024]** | −1.6 ▼ | 3.3 | −1.0 ▼ | #11 | [14] |
| Physics enrollments per 100 high-school students | 10.1[2016] | **8.1[2024]** | −2.0 ▼ | 9.4 | −0.4 ▼ | #28 | [14] |
| SAT or ACT takers per 100 high-school students | 19.2[2016] | **10.1[2024]** | −9.0 ▼ | 18.1 | −2.5 ▼ | #45 | [14] |
| Students in gifted and talented programs per 100 students | 3.6[2016] | **2.8[2024]** | −0.8 ▼ | 7.0 | +0.4 ▲ | #35 | [14] |
| **ENROLLMENT AND INSTITUTIONS** | | | | | | | |
| Pell Grant share of undergraduates | 32% | **34%[2025]** | +2.0 pp ▲ | 36% | −0.8 pp ▼ | #31 | [15] |
| Fall enrollment | 755K | **637K[2024]** | −16% ▼ | 19.7M | −1.7% ▼ | #7 | [4] |
| Share of enrollment at public institutions | 54% | **54%[2024]** | −0.4 pp ▼ | 73% | 0.0 pp | #43 | [4] |
| Share of enrollment at for-profit institutions | 4.7% | **1.8%[2024]** | −2.9 pp ▼ | 5.5% | −1.2 pp ▼ | #26 | [4] |
| Share of students enrolled part-time | 29% | **27%[2024]** | −2.1 pp ▼ | 39% | +0.7 pp ▲ | #48 | [4] |
| Share of students who are graduate students | 18% | **23%[2024]** | +4.6 pp ▲ | 17% | +2.0 pp ▲ | #4 | [4] |
| Share of students enrolled exclusively online | 11% | **17%[2024]** | +5.5 pp ▲ | 27% | +12.1 pp ▲ | #46 | [4] |
| **DEGREES AND FIELDS OF STUDY** | | | | | | | |
| Research doctorates awarded | 2.63K | **2.94K[2025]** | +12% ▲ | 59K | +7.3% ▲ | #5 | [7] |
| Science & engineering research doctorates, excluding health | 1.87K | **2.25K[2025]** | +21% ▲ | 44.4K | +14% ▲ | #5 | [7] |
| Share of U.S. research doctorates | 4.8% | **5.0%[2025]** | +0.2 pp ▲ | 100% | 0.0 pp | #5 | [7] |
| Engineering research doctorates | 618 | **732[2025]** | +18% ▲ | 10.9K | +9.9% ▲ | #3 | [7] |
| Mathematics & computer science research doctorates | 216 | **279[2025]** | +29% ▲ | 5.07K | +33% ▲ | #6 | [7] |
| STEM master's degrees per research doctorate | 4.2 | **5.4[2025]** | +1.2 ▲ | 6.4 | +2.1 ▲ | #25 | [4] |
| STEM share of bachelor's | 21% | **27%[2025]** | +6.2 pp ▲ | 24% | +5.3 pp ▲ | #10 | [4] |
| Share of bachelor's in computer & information sciences | 3.4% | **6.9%[2025]** | +3.5 pp ▲ | 6.9% | +3.7 pp ▲ | #18 | [4] |
| Share of bachelor's in engineering & eng. technology | 6.9% | **8.1%[2025]** | +1.1 pp ▲ | 6.8% | +0.7 pp ▲ | #13 | [4] |
| Share of bachelor's in mathematics & statistics | 1.4% | **1.5%[2025]** | +0.1 pp ▲ | 1.2% | 0.0 pp | #8 | [4] |

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## PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

| Measure | 2015 | Latest | Change since 2015 | U.S., latest | U.S. change since 2015 | Rank | Source |
|---|---|---|---|---|---|---|---|
| **DEGREES AND FIELDS OF STUDY (CONTINUED)** | | | | | | | |
| Share of bachelor's in biological & biomedical sciences | 6.0% | **7.3%**[2025] | +1.2 pp ▲ | 6.6% | +0.8 pp ▲ | #15 | [4] |
| Share of bachelor's in physical sciences | 2.2% | **1.9%**[2025] | −0.3 pp ▼ | 1.3% | −0.3 pp ▼ | #4 | [4] |
| Share of bachelor's in health professions | 12% | **13%**[2025] | +0.4 pp ▲ | 12% | +1.1 pp ▲ | #22 | [4] |
| Share of bachelor's in business | 19% | **17%**[2025] | −1.2 pp ▼ | 19% | −0.1 pp ▼ | #39 | [4] |
| STEM share of master's | 18% | **25%**[2025] | +7.0 pp ▲ | 23% | +6.7 pp ▲ | #15 | [4] |
| Share of master's in computer & information sciences | 5.0% | **9.9%**[2025] | +4.9 pp ▲ | 10% | +6.3 pp ▲ | #17 | [4] |
| Share of master's in engineering & eng. technology | 8.0% | **8.7%**[2025] | +0.7 pp ▲ | 6.7% | −0.2 pp ▼ | #8 | [4] |
| Share of master's in mathematics & statistics | 1.0% | **1.6%**[2025] | +0.7 pp ▲ | 1.3% | +0.3 pp ▲ | #11 | [4] |
| Share of master's in biological & biomedical sciences | 2.1% | **2.9%**[2025] | +0.8 pp ▲ | 2.5% | +0.6 pp ▲ | #19 | [4] |
| Share of master's in physical sciences | 0.9% | **0.7%**[2025] | −0.1 pp ▼ | 0.8% | −0.1 pp ▼ | #28 | [4] |
| Share of master's in health professions | 17% | **17%**[2025] | +0.5 pp ▲ | 15% | +1.7 pp ▲ | #13 | [4] |
| Share of master's in business | 22% | **19%**[2025] | −3.0 pp ▼ | 21% | −3.2 pp ▼ | #32 | [4] |
| STEM share of research doctorates | 50% | **50%**[2025] | +0.3 pp ▲ | 39% | −2.2 pp ▼ | #10 | [4] |
| Share of research doctorates in computer & information sciences | 4.9% | **6.3%**[2025] | +1.4 pp ▲ | 4.2% | +1.4 pp ▲ | #8 | [4] |
| Share of research doctorates in engineering & eng. technology | 20% | **21%**[2025] | +0.5 pp ▲ | 14% | −1.0 pp ▼ | #5 | [4] |
| Share of research doctorates in mathematics & statistics | 2.7% | **2.8%**[2025] | 0.0 pp | 2.4% | −0.2 pp ▼ | #20 | [4] |
| Share of research doctorates in biological & biomedical sciences | 13% | **13%**[2025] | +0.1 pp ▲ | 10% | −1.1 pp ▼ | #18 | [4] |
| Share of research doctorates in physical sciences | 8.2% | **7.0%**[2025] | −1.2 pp ▼ | 7.1% | −1.1 pp ▼ | #33 | [4] |
| Share of research doctorates in health professions | 8.1% | **6.9%**[2025] | −1.2 pp ▼ | 12% | +3.5 pp ▲ | #34 | [4] |
| Share of research doctorates in business | 2.8% | **3.3%**[2025] | +0.5 pp ▲ | 5.6% | +1.3 pp ▲ | #28 | [4] |
| Bachelor's degrees per 100 undergraduates | 14.7 | **16.5**[2025] | +1.8 ▲ | 12.3 | +1.4 ▲ | #6 | [4] |
| **INTERNATIONAL STUDENTS** | | | | | | | |
| First-time students from outside the U.S. | 3.5%[2016] | **3.4%**[2024] | −0.1 pp ▼ | 3.1% | +0.1 pp ▲ | #16 | [4] |
| STEM master's degrees earned by international students | 3.1K | **5.51K**[2025] | +78% ▲ | 117K | 2.18× ▲ | #7 | [4] |
| Share of all U.S. international STEM master's degrees | 5.8% | **4.7%**[2025] | −1.1 pp ▼ | 100% | 0.0 pp | #7 | [4] |

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## PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

| Measure | 2015 | Latest | Change since 2015 | U.S., latest | U.S. change since 2015 | Rank | Source |
|---|---|---|---|---|---|---|---|
| **INTERNATIONAL STUDENTS (CONTINUED)** | | | | | | | |
| International share of undergraduate enrollment | 3.9% | **3.9%**[2024] | 0.0 pp | 3.0% | −0.3 pp ▼ | #9 | [4] |
| International share of graduate enrollment | 16% | **18%**[2024] | +2.1 pp ▲ | 17% | +2.8 pp ▲ | #17 | [4] |
| International share of all bachelor's | 3.6% | **4.8%**[2025] | +1.2 pp ▲ | 4.1% | +0.1 pp ▲ | #10 | [4] |
| International share of STEM bachelor's | 6.1% | **8.3%**[2025] | +2.2 pp ▲ | 6.6% | +0.8 pp ▲ | #6 | [4] |
| International share of all master's | 16% | **24%**[2025] | +7.2 pp ▲ | 22% | +6.2 pp ▲ | #12 | [4] |
| International share of STEM master's | 47% | **54%**[2025] | +6.9 pp ▲ | 52% | +9.1 pp ▲ | #14 | [4] |
| International share of computer & information sciences master's | 63% | **62%**[2025] | −1.1 pp ▼ | 65% | +10.1 pp ▲ | #27 | [4] |
| International share of engineering & eng. technology master's | 49% | **56%**[2025] | +7.6 pp ▲ | 50% | +0.9 pp ▲ | #9 | [4] |
| International share of mathematics & statistics master's | 33% | **56%**[2025] | +23.5 pp ▲ | 55% | +6.1 pp ▲ | #13 | [4] |
| International share of biological & biomedical sciences master's | 23% | **19%**[2025] | −4.2 pp ▼ | 23% | +6.2 pp ▲ | #22 | [4] |
| International share of physical sciences master's | 25% | **35%**[2025] | +9.9 pp ▲ | 33% | +3.6 pp ▲ | #18 | [4] |
| International share of all research doctorates | 31% | **33%**[2025] | +1.8 pp ▲ | 25% | −1.0 pp ▼ | #9 | [4] |
| International share of STEM research doctorates | 43% | **45%**[2025] | +1.3 pp ▲ | 43% | −0.5 pp ▼ | #18 | [4] |
| International share of computer & information sciences research doctorates | 49% | **65%**[2025] | +15.9 pp ▲ | 56% | +2.0 pp ▲ | #7 | [4] |
| International share of engineering & eng. technology research doctorates | 55% | **54%**[2025] | −0.4 pp ▼ | 55% | −1.0 pp ▼ | #23 | [4] |
| International share of mathematics & statistics research doctorates | 59% | **50%**[2025] | −9.1 pp ▼ | 53% | +2.9 pp ▲ | #18 | [4] |
| International share of biological & biomedical sciences research doctorates | 25% | **21%**[2025] | −3.7 pp ▼ | 23% | −4.3 pp ▼ | #29 | [4] |
| International share of physical sciences research doctorates | 39% | **42%**[2025] | +3.1 pp ▲ | 38% | −1.1 pp ▼ | #17 | [4] |
| Active F-1/M-1 students, all levels | 52.4K | **48.6K**[2026] | −7.3% ▼ | 1.19M | −0.1% ▼ | #7 | [20] |
| Share of U.S. active F-1/M-1 students | 4.4% | **4.1%**[2026] | −0.3 pp ▼ | 100% | 0.0 pp | #7 | [20] |
| Share of F-1/M-1 students from India | 15% | **22%**[2026] | +6.9 pp ▲ | 27% | +11.4 pp ▲ | #24 | [20] |
| Share of F-1/M-1 students from China | 39% | **27%**[2026] | −11.7 pp ▼ | 18% | −11.7 pp ▼ | #5 | [20] |
| Share from the two largest countries of origin | 54% | **49%**[2026] | −4.8 pp ▼ | 45% | −0.3 pp ▼ | #15 | [20] |

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## PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

| Measure | 2015 | Latest | Change since 2015 | U.S., latest | U.S. change since 2015 | Rank | Source |
|---|---|---|---|---|---|---|---|
| **INTERNATIONAL STUDENTS (CONTINUED)** | | | | | | | |
| Countries of citizenship represented | 185 | **186**[2026] | +0.5% ▲ | 219 | −2.7% ▼ | #6 | [20] |
| Active STEM F-1/M-1 students | 24.5K | **31.1K**[2026] | +27% ▲ | 749K | +61% ▲ | #7 | [20] |
| Active STEM master's students | 10.8K | **14.6K**[2026] | +36% ▲ | 393K | +90% ▲ | #7 | [20] |
| Active STEM doctoral students | 4.99K | **7.08K**[2026] | +42% ▲ | 158K | +61% ▲ | #6 | [20] |
| Share of STEM master's students from India | 43% | **46%**[2026] | +3.2 pp ▲ | 58% | −1.8 pp ▼ | #37 | [20] |
| U.S. students studying abroad for credit, from the state's institutions | 21K[2017] | **15.8K**[2024] | −25% ▼ | – | – | #4 | [1] |
| Economic contribution of international students | $2.1B[2018] | **$2.1B**[2024] | +0.0% | $42.9B | +4.6% ▲ | #6 | [2] |
| Jobs supported by international students' spending | 27.5K[2018] | **20.2K**[2024] | −27% ▼ | 356K | −22% ▼ | #6 | [2] |
| Share of U.S. student and exchange-visitor admissions, by destination | 3.2% | **3.4%**[2024] | +0.3 pp ▲ | 100% | 0.0 pp | #8 | [16] |
| Share of U.S. F-1 student admissions, by destination | 3.2% | **3.4%**[2024] | +0.2 pp ▲ | 100% | 0.0 pp | #8 | [16] |
| Share of U.S. J-1 exchange-visitor admissions, by destination | 3.2% | **3.9%**[2024] | +0.8 pp ▲ | 100% | 0.0 pp | #7 | [16] |
| J-1 exchange visitors starting a program in the state | 12.6K | **13.1K**[2025] | +4.4% ▲ | 275K | −14% ▼ | #4 | [18] |
| **GRADUATE EDUCATION AND RESEARCH** | | | | | | | |
| Graduate students in science & engineering | 26.4K | **28.4K**[2016] | +7.8% ▲ | 620K | +0.4% ▲ | #6 | [9] |
| S&E graduate students on temporary visas | 10.7K | **12.5K**[2016] | +17% ▲ | 241K | +2.3% ▲ | #5 | [9] |
| Temporary-visa share of S&E graduate students | 41% | **44%**[2016] | +3.6 pp ▲ | 39% | +0.7 pp ▲ | #10 | [9] |
| Temporary-visa share of engineering graduate students | 60% | **60%**[2016] | −0.3 pp ▼ | 57% | 0.0 pp | #12 | [9] |
| S&E postdoctoral appointees | 1.85K | **1.99K**[2016] | +8.0% ▲ | 45.7K | +1.0% ▲ | #5 | [9] |
| Temporary-visa share of S&E postdocs | 59% | **58%**[2016] | −0.5 pp ▼ | 56% | −0.9 pp ▼ | #18 | [9] |
| University R&D spending | $3.36B | **$6.45B**[2024] | +92% ▲ | $118B | +71% ▲ | #5 | [6] |
| Share of U.S. university R&D | 4.9% | **5.5%**[2024] | +0.6 pp ▲ | 100% | 0.0 pp | #5 | [6] |
| Federally funded share of university R&D | 65% | **56%**[2024] | −8.9 pp ▼ | 55% | −0.2 pp ▼ | #17 | [6] |
| Business-funded share of university R&D | 5.9% | **5.8%**[2024] | −0.1 pp ▼ | 5.4% | −0.4 pp ▼ | #12 | [6] |
| State- and locally funded share of university R&D | 2.4% | **2.0%**[2024] | −0.3 pp ▼ | 5.2% | −0.4 pp ▼ | #44 | [6] |
| University R&D in mathematics and computer science | $220M | **$476M**[2024] | 2.16× ▲ | $5.12B | +97% ▲ | #2 | [6] |

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|---|---|---|---|---|---|---|---|
| **GRADUATE EDUCATION AND RESEARCH (CONTINUED)** | | | | | | | |
| University R&D in engineering | $543M | **$952M**[2024] | +75% ▲ | $19.3B | +74% ▲ | #7 | [6] |
| University R&D personnel, headcount | 51.8K | **51.4K**[2024] | −0.9% ▼ | 1.09M | +20% ▲ | #4 | [6] |
| Federal R&D obligations to the state | $3.71B | **$5.15B**[2024] | +39% ▲ | $193B | +53% ▲ | #12 | [8] |
| Federal R&D obligations to universities in the state | $1.66B | **$2.56B**[2024] | +54% ▲ | $44.4B | +64% ▲ | #5 | [8] |
| Defense R&D obligations to the state | $1.24B | **$1.58B**[2024] | +27% ▲ | $97.2B | +60% ▲ | #13 | [8] |
| Health (HHS/NIH) R&D obligations to the state | $1.52B | **$2.26B**[2024] | +48% ▲ | $48.5B | +62% ▲ | #5 | [8] |
| NSF, NIH, NASA and DOE Science grants performed in the state | $1.82B | **$2.57B**[2025] | +42% ▲ | $45B | +47% ▲ | #4 | [19] |
| Federal R&D contracts performed in the state | $630M | **$1.6B**[2026] | 2.54× ▲ | $59.3B | +59% ▲ | #9 | [19] |
| **COLLEGE COSTS AND OUTCOMES** | | | | | | | |
| Median earnings 10 years after entering college | $45K[2020] | **$71.8K**[2022] | +59% ▲ | $47.5K | +11% ▲ | #1 | [15] |
| Median earnings 1 year after completing | – | **$50.4K**[2025] | – | $47.5K | – | #16 | [15] |
| Median debt of completers | $26K | **$23.3K**[2024] | −11% ▼ | $17.6K | −4.6% ▼ | #2 | [15] |
| STEM bachelor's graduates: median earnings 1 year out | – | **$68.2K**[2025] | – | $68.2K | – | #22 | [15] |
| Non-STEM bachelor's graduates: median earnings 1 year out | – | **$49.4K**[2025] | – | $51.9K | – | #31 | [15] |
| Bachelor's graduates: median earnings 5 years out | – | **$69.5K**[2025] | – | $68.1K | – | #15 | [15] |
| STEM master's graduates: median earnings 1 year out | – | **$103K**[2025] | – | $91.6K | – | #5 | [15] |
| Six-year bachelor's graduation rate, 4-year institutions | 67% | **68%**[2024] | +0.9 pp ▲ | 65% | +5.3 pp ▲ | #15 | [4] |
| Completion within 150% of normal time, 2-year institutions | 37% | **42%**[2024] | +5.4 pp ▲ | 41% | +12.1 pp ▲ | #22 | [4] |
| In-state tuition & fees, public 4-year | $17.8K | **$17K**[2023] | −4.3% ▼ | $9.71K | −14% ▼ | #3 | [4] |
| State & local appropriations per FTE, public institutions | $5.97K | **$7.14K**[2024] | +20% ▲ | $11.4K | +22% ▲ | #46 | [4] |
| Tuition share of core revenue, public institutions | 76% | **72%**[2024] | −4.0 pp ▼ | 43% | −5.9 pp ▼ | #3 | [4] |
| Average full-time faculty salary (9/10-month) | $119K[2016] | **$109K**[2024] | −8.2% ▼ | $106K | −4.6% ▼ | #14 | [4] |
| **WHERE STUDENTS AND GRADUATES GO** | | | | | | | |
| International graduates on OPT working in the state where they studied | 29% | **26%**[2016] | −2.6 pp ▼ | 44% | −1.8 pp ▼ | #38 | [21] |
| OPT retention, bachelor's graduates | 37% | **30%**[2016] | −6.9 pp ▼ | 58% | +0.3 pp ▲ | #44 | [21] |

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| Measure | 2015 | Latest | Change since 2015 | U.S., latest | U.S. change since 2015 | Rank | Source |
|---|---|---|---|---|---|---|---|
| **WHERE STUDENTS AND GRADUATES GO (CONTINUED)** | | | | | | | |
| OPT retention, master's graduates | 25% | **24%**[2016] | −1.4 pp ▼ | 39% | −2.5 pp ▼ | #35 | [21] |
| OPT retention, STEM graduates | 25% | **22%**[2016] | −2.3 pp ▼ | 37% | −1.9 pp ▼ | #35 | [21] |
| OPT workers employed in the state per OPT graduate of its schools | 0.58 | **0.58**[2016] | 0.0 | 1.00 | 0.0 | #37 | [21] |
| STEM OPT extensions approved, by state of the school | 1.56K | **1.95K**[2016] | +25% ▲ | 41.2K | +51% ▲ | #5 | [21] |
| Bachelor's graduates working in the state 1 year after graduating (PSEO) | 55%[2016] | **55%**[2019] | +0.2 pp ▲ | 72% | −0.1 pp ▼ | #27 | [11] |
| Bachelor's graduates working in the state 5 years after graduating (PSEO) | – | **51%**[2016] | – | 65% | – | #26 | [11] |
| Bachelor's graduates working in the state 10 years after graduating (PSEO) | – | **51%**[2010] | – | 64% | – | #26 | [11] |
| STEM bachelor's graduates working in the state 1 year out (PSEO) | 52%[2016] | **55%**[2019] | +2.7 pp ▲ | 67% | +1.5 pp ▲ | #23 | [11] |
| Master's graduates working in the state 1 year out (PSEO) | – | **48%**[2016] | – | 66% | – | #31 | [11] |
| Doctoral graduates working in the state 1 year out (PSEO) | – | **40%**[2016] | – | 49% | – | #27 | [11] |
| Bachelor's graduates' median earnings 1 year out (PSEO) | $54.8K[2016] | **$55K**[2019] | +0.2% ▲ | $48.1K | +2.3% ▲ | #2 | [11] |
| Bachelor's graduates' median earnings 5 years out (PSEO) | – | **$79.5K**[2016] | – | $67.3K | – | #2 | [11] |
| STEM bachelor's graduates' median earnings 1 year out (PSEO) | $65.4K[2016] | **$64.9K**[2019] | −0.8% ▼ | $59.3K | +0.1% ▲ | #3 | [11] |
| Net domestic migration of bachelor's-plus adults, per 1,000 residents | −4.9 | **−2.6**[2025] | +2.3 ▲ | – | – | #34 | [12] |
| Net domestic migration of graduate-degree holders | −10.9K | **−10.3K**[2025] | – ▲ | – | – | #50 | [12] |
| Bachelor's-plus adults who moved in from another state | 2.4% | **2.2%**[2025] | −0.2 pp ▼ | – | – | #38 | [12] |
| Bachelor's-plus adults who moved out to another state | 2.9% | **2.4%**[2025] | −0.5 pp ▼ | – | – | #34 | [12] |
| Bachelor's-plus adults who arrived from abroad in the past year | 16.8K | **18K**[2025] | +7.1% ▲ | – | – | #11 | [12] |
| First-time students who are in-state residents | 71%[2016] | **68%**[2024] | −2.4 pp ▼ | – | – | #31 | [4] |
| Net student migration (per 100 resident first-time students) | +11[2016] | **+8**[2024] | −2.9 ▼ | – | – | #21 | [4] |
| Recent high school graduates who go to college out of state | 20%[2016] | **27%**[2022] | +7.6 pp ▲ | 22% | +2.5 pp ▲ | #19 | [3] |
| **WORK VISAS AND GREEN CARDS** | | | | | | | |
| Certified H-1B positions, by worksite | 63.8K | **20K**[2025] | −69% ▼ | 914K | −24% ▼ | #12 | [17] |

Superscript years mark the year observed; changes since 2015 start from 2015 or the first later year with data. Rank is among the 50 states and DC (1 = highest). "pp" is percentage points. Dollars are 2024 dollars (CPI-U) unless the measure is labeled nominal. Counts and dollars are rounded to three significant figures (K, M, B). Arrow colors show direction only. Source numbers refer to the list of sources that follows the tables.

## PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

| Measure | 2015 | Latest | Change since 2015 | U.S., latest | U.S. change since 2015 | Rank | Source |
|---|---|---|---|---|---|---|---|
| **WORK VISAS AND GREEN CARDS (CONTINUED)** | | | | | | | |
| Share of U.S. certified H-1B positions | 5.3% | **2.2%**$^{2025}$ | −3.1 pp ▼ | 100% | 0.0 pp | #12 | [17] |
| STEM share of certified H-1B positions | 84% | **79%**$^{2025}$ | −4.5 pp ▼ | 76% | −7.8 pp ▼ | #11 | [17] |
| Median offered H-1B wage, nominal dollars | $69.5K | **$103K**$^{2025}$ | +49% ▲ | $119K | +66% ▲ | #27 | [17] |
| H-1B positions at H-1B-dependent employers | 31% | **24%**$^{2025}$ | −6.9 pp ▼ | 22% | −25.7 pp ▼ | #17 | [17] |
| Green-card labor certifications | 1.8K | **3.18K**$^{2025}$ | +77% ▲ | 138K | +75% ▲ | #13 | [17] |
| Share of PERM certifications for citizens of India | 64% | **45%**$^{2024}$ | −18.7 pp ▼ | 47% | −10.6 pp ▼ | #19 | [17] |
| PERM certifications for workers on F-1 student visas | 2.4% | **9.9%**$^{2024}$ | +7.6 pp ▲ | 8.7% | +4.9 pp ▲ | #24 | [17] |
| PERM workers with a master's or doctorate | 53% | **44%**$^{2024}$ | −8.5 pp ▼ | 47% | −4.9 pp ▼ | #20 | [17] |
| STEM share of PERM certifications | 71% | **51%**$^{2025}$ | −20.3 pp ▼ | 49% | −25.8 pp ▼ | #14 | [17] |
| Median offered PERM wage, nominal dollars | $90.5K | **$94.7K**$^{2025}$ | +4.7% ▲ | $100K | +9.9% ▲ | #20 | [17] |
| New lawful permanent residents (green cards), by state of residence | 25K | **34.9K**$^{2024}$ | +40% ▲ | 1.36M | +30% ▲ | #10 | [16] |
| Share of U.S. new green-card holders | 2.4% | **2.6%**$^{2024}$ | +0.2 pp ▲ | 100% | 0.0 pp | #10 | [16] |
| People naturalized as U.S. citizens, by state of residence | 16.6K | **22.2K**$^{2024}$ | +34% ▲ | 819K | +12% ▲ | #11 | [16] |
| Share of U.S. naturalizations | 2.3% | **2.7%**$^{2024}$ | +0.5 pp ▲ | 100% | 0.0 pp | #11 | [16] |
| Share of U.S. temporary-worker admissions, by destination | 1.5% | **1.3%**$^{2024}$ | −0.2 pp ▼ | 100% | 0.0 pp | #13 | [16] |
| Share of U.S. H-1B worker admissions, by destination | 2.3% | **1.9%**$^{2024}$ | −0.4 pp ▼ | 100% | 0.0 pp | #12 | [16] |
| Share of U.S. L-1 intracompany-transferee admissions, by destination | 2.0% | **1.8%**$^{2024}$ | −0.2 pp ▼ | 100% | 0.0 pp | #12 | [16] |
| Share of U.S. O-1 extraordinary-ability worker admissions, by destination | 1.0% | **1.1%**$^{2024}$ | +0.1 pp ▲ | 100% | 0.0 pp | #12 | [16] |
| New H-1B approvals (initial), by employer state | 3.38K | **3.6K**$^{2026}$ | +6.7% ▲ | 127K | +14% ▲ | #9 | [13] |
| Continuing H-1B approvals, by employer state | 5.76K | **8.55K**$^{2026}$ | +48% ▲ | 268K | +52% ▲ | #8 | [13] |
| Share of U.S. new H-1B approvals | 3.0% | **2.8%**$^{2026}$ | −0.2 pp ▼ | 100% | 0.0 pp | #9 | [13] |
| Denial rate, new H-1B petitions | 10% | **2.7%**$^{2026}$ | −7.8 pp ▼ | 3.4% | −3.0 pp ▼ | #41 | [13] |
| New H-1B approvals at colleges and schools (NAICS 61) | 17% | **13%**$^{2026}$ | −4.3 pp ▼ | 8.6% | −1.7 pp ▼ | #30 | [13] |
| New H-1B approvals in professional, scientific & technical services | 37% | **50%**$^{2026}$ | +13.4 pp ▲ | 44% | −12.5 pp ▼ | #9 | [13] |

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## PENNSYLVANIA – BY THE NUMBERS (CONTINUED)

| Measure | 2015 | Latest | Change since 2015 | U.S., latest | U.S. change since 2015 | Rank | Source |
|---|---|---|---|---|---|---|---|
| **WORK VISAS AND GREEN CARDS (CONTINUED)** | | | | | | | |
| Five largest employers' share of new H-1B approvals | 32% | **26%[2026]** | −6.7 pp ▼ | 9.1% | −6.7 pp ▼ | #35 | [13] |
| **STEM WORKFORCE AND INNOVATION** | | | | | | | |
| Workers in STEM occupations | 344K | **521K[2024]** | +51% ▲ | 13.7M | +51% ▲ | #5 | [12] |
| STEM workers as a share of all workers | 5.7% | **8.0%[2024]** | +2.4 pp ▲ | 8.1% | +2.1 pp ▲ | #20 | [12] |
| Foreign-born share of STEM workers | 14% | **16%[2024]** | +2.2 pp ▲ | 24% | +1.6 pp ▲ | #21 | [12] |
| Foreign-born share of STEM workers with a graduate degree | 29% | **29%[2024]** | −0.2 pp ▼ | 41% | +2.2 pp ▲ | #23 | [12] |
| Foreign-born share of STEM workers with a doctorate | 46% | **29%[2024]** | −17.6 pp ▼ | 47% | +0.9 pp ▲ | #19 | [12] |
| Noncitizen share of STEM workers | 7.2% | **7.5%[2024]** | +0.3 pp ▲ | 11% | +0.7 pp ▲ | #22 | [12] |
| STEM workers born abroad who arrived in the last 10 years | 5.0% | **4.6%[2024]** | −0.4 pp ▼ | 7.5% | +0.3 pp ▲ | #30 | [12] |
| Share of foreign-born STEM workers born in India | 35% | **35%[2024]** | −0.1 pp ▼ | 28% | +0.7 pp ▲ | #7 | [12] |
| Foreign-born share of residents | 6.5% | **8.4%[2025]** | +1.9 pp ▲ | 15% | +1.3 pp ▲ | #26 | [12] |
| Foreign-born share of graduate-degree holders, 25+ | 13% | **15%[2025]** | +2.6 pp ▲ | 21% | +2.8 pp ▲ | #20 | [12] |
| STEM jobs, employer survey | 310K | **377K[2025]** | +22% ▲ | 10.4M | +26% ▲ | #7 | [10] |
| STEM jobs concentration vs the U.S., location quotient | 0.91 | **0.93[2025]** | 0.0 | 1.00 | 0.0 | #22 | [10] |
| Mean annual STEM wage, nominal | $79.8K | **$109K[2025]** | +37% ▲ | $120K | +38% ▲ | #22 | [10] |
| Median software developer wage, nominal | $96.5K[2019] | **$127K[2025]** | +31% ▲ | $136K | +26% ▲ | #26 | [10] |
| Software developer jobs | 32.5K | **55.9K[2025]** | +72% ▲ | 1.69M | +48% ▲ | #9 | [10] |
| Mean annual wage, all occupations, nominal | $46.5K | **$65.7K[2025]** | +41% ▲ | $69.8K | +44% ▲ | #25 | [10] |
| Patents granted to assignees in the state | 2.77K | **3.2K[2025]** | +16% ▲ | 135K | −0.6% ▼ | #12 | [22] |
| Share of U.S.-assignee patents | 2.0% | **2.4%[2025]** | +0.3 pp ▲ | 100% | 0.0 pp | #12 | [22] |
| Share of the state's patents in digital technologies | 16% | **28%[2025]** | +12.5 pp ▲ | 43% | +1.6 pp ▲ | #17 | [22] |
| Share of the state's patents in life sciences | 27% | **28%[2025]** | +1.6 pp ▲ | 17% | +0.4 pp ▲ | #10 | [22] |
| Share of the state's patents in semiconductors | 1.7% | **0.8%[2025]** | −0.9 pp ▼ | 2.9% | −1.7 pp ▼ | #23 | [22] |
| Women's share of inventors on the state's patents | 10% | **12%[2025]** | +1.2 pp ▲ | 14% | +3.4 pp ▲ | #25 | [22] |

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# Sources


**[1]** Institute of International Education. *Open Doors Report on International Educational Exchange: State Fact Sheets*. Editions 2020–2026. New York: Institute of International Education. https://opendoorsdata.org/. Accessed November 2026.

**[2]** NAFSA: Association of International Educators. *NAFSA International Student Economic Value Tool*. Academic years 2018–19 to 2024–25. Washington, DC: NAFSA. https://www.nafsa.org/isev. Accessed November 2026.

**[3]** National Center for Education Statistics. *Digest of Education Statistics*. Editions 1995–2025. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://nces.ed.gov/programs/digest/. Accessed November 2026.

**[4]** National Center for Education Statistics. *Integrated Postsecondary Education Data System (IPEDS)*. Survey files, 1980–2025. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://nces.ed.gov/ipeds/. Accessed November 2026.

**[5]** National Center for Education Statistics. *National Assessment of Educational Progress (NAEP)*. NAEP Data Service, 1990–2024. Washington, DC: U.S. Department of Education, Institute of Education Sciences. https://www.nationsreportcard.gov/. Accessed November 2026.

**[6]** National Center for Science and Engineering Statistics. *Higher Education Research and Development Survey*. Data tables, FY2003–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/higher-education-research-development. Accessed November 2026.

**[7]** National Center for Science and Engineering Statistics. *Survey of Earned Doctorates*. Data tables, 1994–2025. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/earned-doctorates. Accessed November 2026.

**[8]** National Center for Science and Engineering Statistics. *Survey of Federal Funds for Research and Development*. Data tables, FY1985–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/federal-funds-research-development. Accessed November 2026.

**[9]** National Center for Science and Engineering Statistics. *Survey of Graduate Students and Postdoctorates in Science and Engineering*. Data tables, 1994–2024. Alexandria, VA: National Science Foundation. https://ncses.nsf.gov/surveys/graduate-students-postdoctorates-s-e. Accessed November 2026.

**[10]** U.S. Bureau of Labor Statistics. *Occupational Employment and Wage Statistics*. State estimates, 1999–2025. Washington, DC: U.S. Department of Labor. https://www.bls.gov/oes/. Accessed November 2026.

**[11]** U.S. Census Bureau, Center for Economic Studies. *Post-Secondary Employment Outcomes (PSEO)*. Release R2025. Washington, DC: U.S. Department of Commerce. https://lehd.ces.census.gov/data/pseo_experimental.html. Accessed November 2026.

**[12]** U.S. Census Bureau. *American Community Survey*. 1-year estimates and Public Use Microdata Sample, 2010–2025. Washington, DC: U.S. Department of Commerce. https://data.census.gov/. Accessed November 2026.

**[13]** U.S. Citizenship and Immigration Services. *H-1B Employer Data Hub*. Fiscal years 2009–2026. Washington, DC: U.S. Department of Homeland Security. https://www.uscis.gov/tools/reports-and-studies/h-1b-employer-data-hub. Accessed November 2026.

**[14]** U.S. Department of Education, Office for Civil Rights. *Civil Rights Data Collection*. School years 1980–81 to 2023–24. Washington, DC: U.S. Department of Education. https://civilrightsdata.ed.gov/. Accessed November 2026.

**[15]** U.S. Department of Education. *College Scorecard*. Institution and field-of-study data, 1996–2025. Washington, DC: U.S. Department of Education. https://collegescorecard.ed.gov/data/. Accessed November 2026.

**[16]** U.S. Department of Homeland Security, Office of Homeland Security Statistics. *Yearbook of Immigration Statistics*. Fiscal years 1996–2024. Washington, DC: U.S. Department of Homeland Security. https://ohss.dhs.gov/topics/immigration/yearbook. Accessed November 2026.

**[17]** U.S. Department of Labor, Employment and Training Administration, Office of Foreign Labor Certification. *LCA and PERM Disclosure Data*. Fiscal years 2008–2026. Washington, DC: U.S. Department of Labor. https://www.dol.gov/agencies/eta/foreign-labor/performance. Accessed November 2026.

**[18]** U.S. Department of State, Bureau of Educational and Cultural Affairs. *BridgeUSA: Exchange Visitor Program Participants by State*. 2015–2025. Washington, DC: U.S. Department of State. https://j1visa.state.gov/. Accessed November 2026.

**[19]** U.S. Department of the Treasury, Bureau of the Fiscal Service. *USAspending.gov*. Award data, FY2008–2026. Washington, DC: U.S. Department of the Treasury. https://www.usaspending.gov/. Accessed November 2026.

**[20]** U.S. Immigration and Customs Enforcement, Student and Exchange Visitor Program. *SEVIS by the Numbers: Data Mapping Tool*. Monthly data, 2014–2026. Washington, DC: U.S. Department of Homeland Security. https://studyinthestates.dhs.gov/sevis-data-mapping-tool. Accessed November 2026.

**[21]** U.S. Immigration and Customs Enforcement. *Student and Exchange Visitor Information System (SEVIS) Data*. Freedom of Information Act releases, 2008–2016. Washington, DC: U.S. Department of Homeland Security. https://www.ice.gov/foia/library. Accessed November 2026.

**[22]** U.S. Patent and Trademark Office. *PatentsView*. Annualized patent data, grant years 1976–2025. Alexandria, VA: U.S. Patent and Trademark Office. https://patentsview.org/. Accessed November 2026.